MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

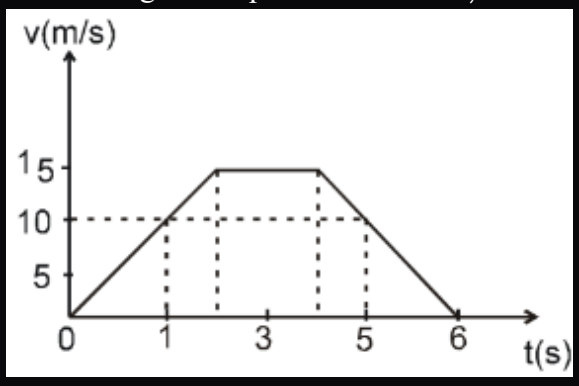
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

A particle of mass 100 gram moves on a straight line. The variation of speed with time is shown in figure. Find the force acting on the particle at $t = 1, 3$ and 5 seconds:-



(A) $2\ \text{N}, 0, -1\ \text{N}$

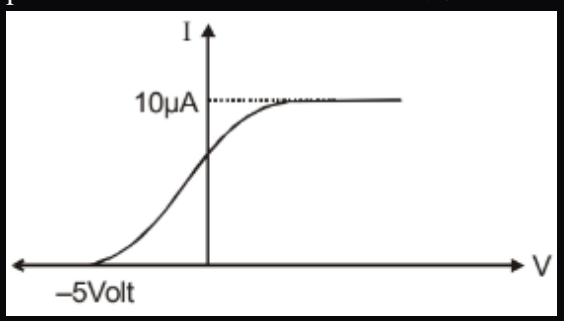
(B) $2\ \text{N}, 0, -2\ \text{N}$

(C) $1\ \text{N}, 0, -1\ \text{N}$

(D) none of these

Q.2

In the photoelectric experiment, if we use a monochromatic light, the $\text{I} - \text{V}$ curve is as shown. If work function of the metal is 2eV, estimate the power of light used. (Assume efficiency of photo emission $= 10^{-3}\%$, i.e. number of photoelectrons emitted are $10^{-3}\%$ of number of photons incident on metal.)



(A) $2\ \mathrm{W}$

(B) $5\ \mathrm{W}$

(C) $7\ \mathrm{W}$

(D) $10\ \mathrm{W}$

Q.3

Force is given by the expression, $F = A\cos(Bx) + C\cos(Dt)$, where x is displacement and t is time. The dimension of $\left(\frac{D}{B}\right)$ is same as that of

(A) velocity

(B) velocity gradient

(C) angular velocity

(D) angular momentum

Q.4

Spacing between two successive nodes in a standing wave on a string is x. If frequency of the standing wave is kept unchanged but tension in the string is doubled, then new spacing between successive nodes will become :

(A) $\mathrm{x}/\sqrt{2}$

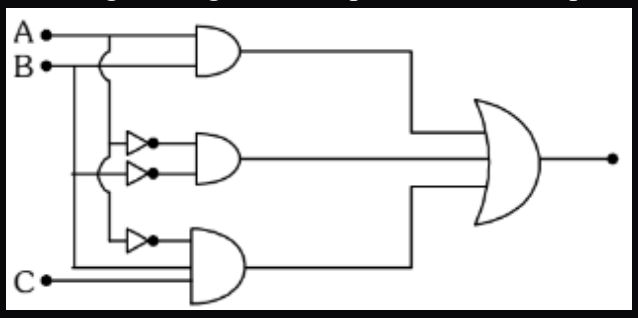
(B) $\sqrt{2}\mathrm{x}$

(C) $x/2$

(D) $2\mathrm{x}$

Q.5

For the given logic circuit; pick the correct option



(A) For $\mathrm{A} = 1,\ \mathrm{B} = 0, \mathrm{C} = 0$; Output $= 1$

(B) For $\mathrm{A} = 0,\ \mathrm{B} = 1, \mathrm{C} = 1$; Output $= 1$

(C) For $\mathrm{A} = 1,\ \mathrm{B} = 1, \mathrm{C} = 0$; Output $= 0$

(D) For $\mathrm{A} = 0,\ \mathrm{B} = 1, \mathrm{C} = 1$; Output $= 0$

Q.6

In a meter bridge, the gaps are enclosed by resistances of 2Ω and 3Ω. The value of shunt to be added to 3Ω resistor to shift the balancing point by $22.5\ \text{cm}$ is

(A) $1\ \Omega$

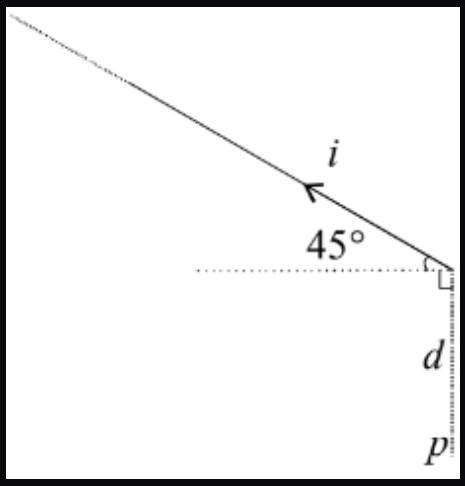
(B) $2\ \Omega$

(C) $2.5\ \Omega$

(D) $5\ \Omega$

Q.7

Find the magnitude of magnetic field at point p due to a semi - infinite wire given below :-



(A) $\frac{\mu_0 i}{4\pi d}(\sqrt{2}-1)$

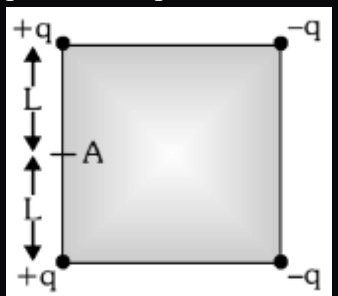
(B) $\frac{\mu_0 i}{4\pi d}(1-\sqrt{2})$

(C) $\frac{\mu_0 i}{4\pi d}\left(1-\frac{1}{\sqrt{2}}\right)$

(D) $\frac{\sqrt{2}\mu_0 i}{4\pi d}$

Q.8

Four electric charges $+q$, $+q$, $-q$, and $-q$ are placed at the corners of a square of side $2\ \text{L}$ (Figure). The electric potential at point A, midway between the two charges $+q$ and $+q$, is



(A) zero

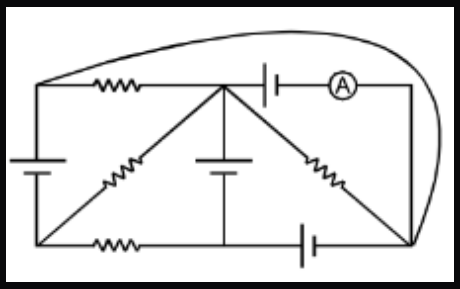
(B) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{\text{L}}(1+\sqrt{3})$

(C) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{\text{L}}\left(1+\frac{1}{\sqrt{5}}\right)$

(D) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{\text{L}}\left(1-\frac{1}{\sqrt{5}}\right)$

Q.9

In the circuit diagram shown, each battery is ideal having an e.m.f. of 1 volt. Each resistor has a resistance of 1Ω. Ammeter (A) has a resistance of 1Ω. Find the reading of the ammeter and the total thermal power produced in the circuit.



(A) $1\ \text{A}, 22\ \text{W}$

(B) $1\ \text{A}, 44\ \text{W}$

(C) $1/2\ \text{A}, 11\ \text{W}$

(D) $2\ \text{A}, 88\ \text{W}$

Q.10

Maximum height reached by a rocket fired with a speed equal to 50% of the escape velocity from earth's surface is :

(A) $R/2$

(B) $16R/9$

(C) $R/3$

(D) $R/8$

Q.11

A soap bubble (surface tension $= T$) is charged to a maximum surface density of charge $= \sigma$. When it is just going to burst. Its radius R is given by:

(A) $R = \frac{\sigma^2}{8\varepsilon_0 T}$

(B) $R = 8\varepsilon_0 \frac{T}{\sigma^2}$

(C) $\mathrm{R} = \frac{\sigma}{\sqrt{8\varepsilon_0\ \mathrm{T}}}$

(D) $R = \frac{\sqrt{8\varepsilon_0\ \mathrm{T}}}{\sigma}$

Q.12

The pitch of a screw guage is $0.5\ \text{mm}$ and there are 100 divisions on its circular scale. The instrument reads +2 divisions when nothing is put in-between its jaws. In measuring the diameter of a wire, there are 8 divisions on the main scale and 83rd divison coincides with the reference line. Then the diameter of the wire is

(A) 4.05 mm

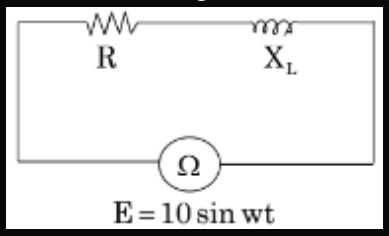
(B) 4.405 mm

(C) 3.5 mm

(D) 1.25 mm

Q.13

An AC circuit having supply voltage E consists of a resistor of resistance 3Ω and an inductor of reactance 4Ω as shown in the figure. The voltage across the inductor at $t = \frac{T}{2}$ is



(A) 2 V

(B) 10 V

(C) Zero

(D) 4.8 V

Q.14

An automatic gun fires 360 bullets per minute with a speed of $360\ \mathrm{km/h}$. If each bullet weighs $20\ \mathrm{g}$, the power of the gun is

(A) 75 W

(B) 150 W

(C) 300 W

(D) 600 W

Q.15

In the visible region of the spectrum the rotation of the plane of polarization is given by $\theta = a + \frac{b}{\lambda^2}$. The optical rotation produced by a particular material is found to be 30° per mm at $\lambda = 5000\tilde{A}\ldots$ and 50° per mm at $\lambda = 4000\tilde{A}\ldots$. The value of constant a will be

(A) $+\frac{50^\circ}{9}$ per mm

(B) $-\frac{50^\circ}{9}$ per mm

(C) $+\frac{9^\circ}{50}$ per mm

(D) $-\frac{9^\circ}{50}$ per mm

Q.16

Let A_n be the area enclosed by the nth orbit in a hydrogen atom. The graph of $\ell n\left(A_n/A_1\right)$ against ℓ n(n)

(A) will not pass through the origin

(B) will have certain points lying on a straight line with slope 4

(C) will be a monotonically increasing non linear curve

(D) will be a circle

Q.17

In a two dimensional motion of a particle, the particle moves from point A, position vector r_1 to point B position vector r_2. If the magnitude of these vector are respectively $r_1 = 3$ and $r_2 = 4$ and the angle they make with the x-axis are $\theta_1 = 75^\circ, \theta_2 = 15^\circ$ respectively, then find the magnitude of the displacement vector. If it is $\sqrt{\mathrm{n}+4}$, find n.

(A) 6

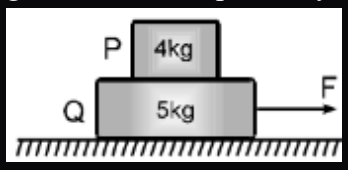
(B) 4

(C) 9

(D) 3

Q.18

In the given figure the coefficient of friction between $4\ \mathrm{kg}$ and $5\ \mathrm{kg}$ blocks is 0.2 and between $5\ \mathrm{kg}$ block and ground is 0.1 respectively. Choose the correct statements :-



(A) Minimum force needed to cause system to move is $17\ \mathrm{N}$

(B) When force is $4\ \mathrm{N}$ static friction at all surfaces is $4\ \mathrm{N}$ to keep system at rest

(C) Maximum acceleration of $4\ \mathrm{kg}$ block is $2\ \mathrm{m/s^2}$

(D) Slipping between $4\ \mathrm{kg}$ and $5\ \mathrm{kg}$ blocks start when F is $> 17\ \mathrm{N}$

Q.19

Which of the following is correct for the molecules of a gas in thermal equilibrium?

(A) All have the same speed

(B) All have different speeds which remain constant

(C) They have a certain constant average speed

(D) They do not collide with one another

Q.20

114. Match the electromagnetic radiations given in List -I with their uses given in List - II.

List - I	List - II
A) X-rays	P) Remote switches
B) UV-rays	Q) Finger prints in forensic Labs
C) Radio waves	R) Crystal structure study
D) IR - rays	S) TV communication system.

(A) $\mathrm{A} \rightarrow \mathrm{Q}, \mathrm{B} \rightarrow \mathrm{R}, \mathrm{C} \rightarrow \mathrm{P}, \mathrm{D} \rightarrow \mathrm{S}$

(B) $\mathrm{A} \rightarrow \mathrm{R}, \mathrm{B} \rightarrow \mathrm{Q}, \mathrm{C} \rightarrow \mathrm{S}, \mathrm{D} \rightarrow \mathrm{P}$

(C) $\mathrm{A} \rightarrow \mathrm{R}, \mathrm{B} \rightarrow \mathrm{S}, \mathrm{C} \rightarrow \mathrm{Q}, \mathrm{D} \rightarrow \mathrm{P}$

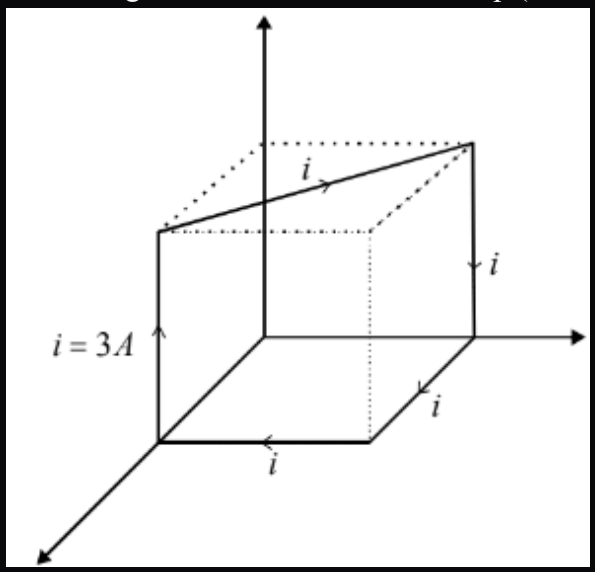
(D) $\mathrm{A} \rightarrow \mathrm{S}, \mathrm{B} \rightarrow \mathrm{R}, \mathrm{C} \rightarrow \mathrm{Q}, \mathrm{D} \rightarrow \mathrm{P}$

Q.21

Two equal point charges are fixed at $x = -a$ and $\mathrm{x} = +\mathrm{a}$ on the x-axis. Another point charge Q is placed at the origin. The change in the electrical potential energy of Q, when it is displaced by a small distance x along the x-axis, is approximately proportional to x^p, find integral value of p.

Q.22

A wire carrying $3\ \mathrm{A}$ current is wrapped around a non - conducting cube of side $1\ \mathrm{m}$ as shown below. The magnitude of net magnetic moment due to the loop (in S.I. unit) is M, find 4M.



Q.23

A boat is moving due east in a region where the earth's magnetic field is $5.0 \times 10^{-5} \mathrm{NA}^{-1}\ \mathrm{m}^{-1}$ due north and horizontal. The boat carries a vertical aerial $3\ \mathrm{m}$ long. If the speed of the boat is $3\ \mathrm{m/s}$. The magnitude of the induced emf in the wire of aerial (in mV) is K, find value of 100K.

Q.24

When a body of mass $1.0\ \mathrm{kg}$ is suspended from a certain light spring hanging vertically, its length increases by $5\ \mathrm{cm}$.

By suspending $2.0\ \mathrm{kg}$ block to the spring and if the block is pulled through $10\ \mathrm{cm}$ and released, the maximum velocity in it in $\mathrm{m/s}$ is : (Acceleration due to gravity $= 10\ \mathrm{m/s^2}$)

Q.25

A ray of light is falling on a glass sphere of $\mu = \sqrt{3}$ such that the incident ray and the emergent ray, when produced, intersect at a point on the surface of the sphere. The angle of incidence in degrees is found to be $\frac{2\pi}{k}$. Find the value of k.

Q.26

On the basis of which physical property separation of different fractions in fractional distillation technique takes place?

(A) Solubility

(B) Freezing point

(C) Boiling point

(D) Melting point

Q.27

Assume that the decomposition of $\mathrm{HNO_3}$ is
$4\mathrm{HNO_3(\ g)} \rightleftharpoons 4\mathrm{NO_2(\ g)} + 2\mathrm{H_2O(g)} + \mathrm{O_2(\ g)}$
and the reaction approaches equilibrium at $400\ \mathrm{K}$ & 30 atm pressure. At equilibrium the partial pressure of $\mathrm{HNO_3}$ is 2 atm. Find K_C at $400\ \mathrm{K}\ (\mathrm{R} = 0.08\ell - \mathrm{atm/K - mol})$

(A) 4

(B) 8

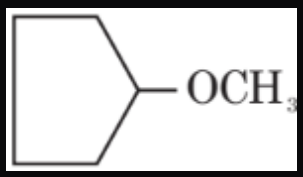
(C) 16

(D) 32

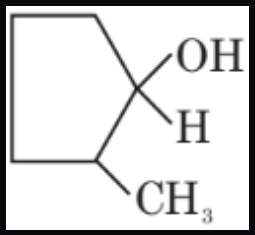
Q.28

An organic compound A (Molecular formula $\mathrm{C_6H_{12}O}$) does not change the colour of acidic dichromate solution Compound A on treatment with $\mathrm{H_2SO_4}$ produces alkene, which on oxidative ozonolysis gives a molecule $(\mathrm{C_6H_{10}O_3})$ which gives positive iodoform test. Find out the structure of ' A '.

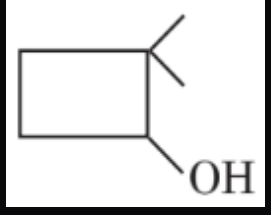
(A)



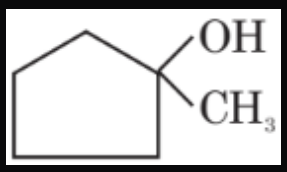
(B)



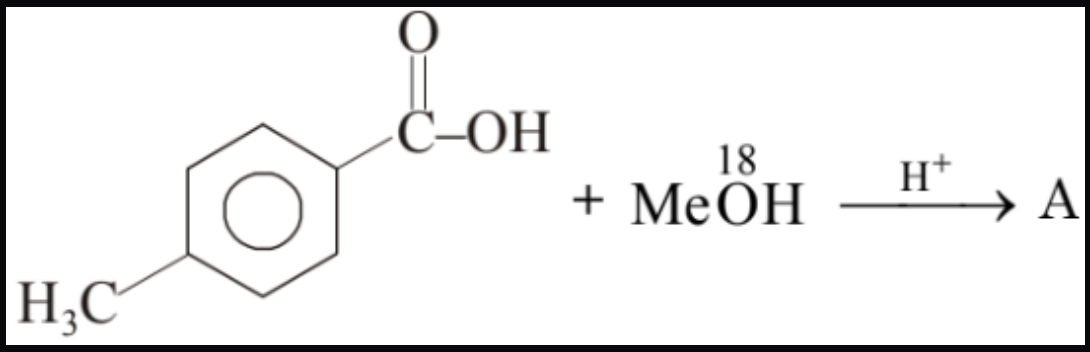
(C)



(D)

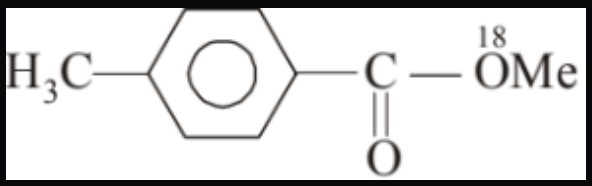


Q.29

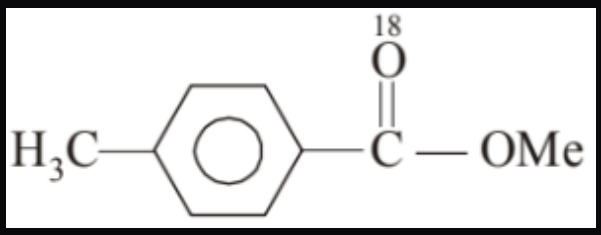


A is

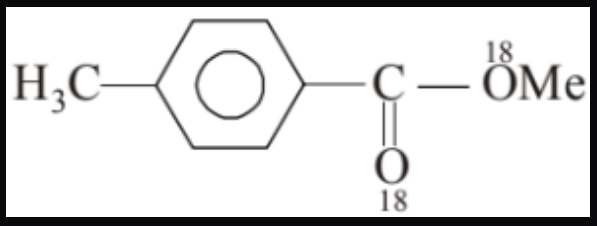
(A)



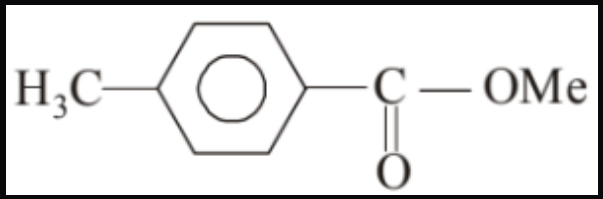
(B)



(C)

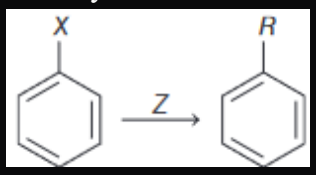


(D)



Q.30

Identify Z in the reaction :



(A) $RCOX/$ Anhydrous $AlCl_3$

(B) $RX/Na/dry\ (C_2H_5)_2\ O$

(C) $Na/$ dry $(C_2H_5)_2\ O$

(D) $RCOX/FeCl_3$

Q.31

In the following sub questions, choose the correct answer from among the following possibilities and select correct code of your answer (Answer of 1, 2, 3 and 4 respectively)

1) The most stable low valent halide (1) $GeCl_2$ (2) $SnCl_2$ (3) $PbCl_2$.

2) A non existing halide (1) $SnCl_4$ (2) $PbCl_4$ (3) PbI_4.

3) A purely acidic oxide (1) PbO_2 (2) SnO_2 (3) SiO_2

4) Thermally most stable hydride (1) NH_3 (2) PH_3 (3) AsH_3.

(A) $3, 2, 1, 3$

(B) $1, 3, 3, 1$

(C) $3, 3, 3, 1$

(D) $1, 1, 1, 3$

Q.32

For a aqueous metal ion d-d transition is observed by using visible light. Which is INCORRECT combination for absorbed light & observed colour.

(A)

Absorbed light	Observe colour
Yellow	Violet

(B)

Absorbed light	Observe colour
Orange	Blue

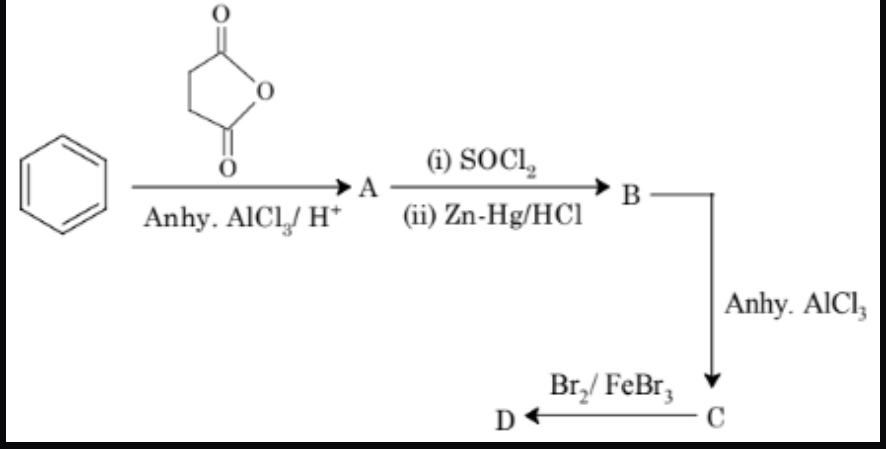
(C)

Absorbed light	Observe colour
Green	Red

(D)

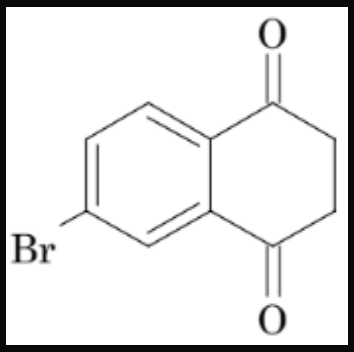
Absorbed light	Observe colour
Blue	Green

Q.33

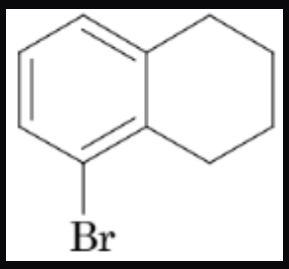


Find D

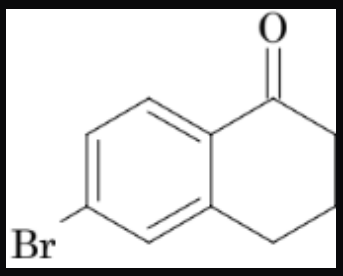
(A)



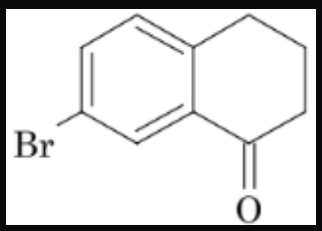
(B)



(C)



(D)



Q.34

The elements with metallic nature in the following are C, Si, Ge, Sn, Pb

(A) Ge, Pb

(B) Ge, Sn

(C) C, Ge

(D) Sn, Pb

Q.35

Which species among the following doesn't show disproportionation reactions?

(A) ClO_2^-

(B) ClO_3^-

(C) ClO^-

(D) ClO_4^-

Q.36

Match the following List - I with List - II

List - I	List - II
P) $Na_2[Zn(OH)_4] + H_2S \rightarrow$	1) Precipitate is formed through non - redox reaction
Q) $Hg\downarrow + HgS\downarrow + \text{aqua - regia} \xrightarrow[\text{(excess)}]{\Delta}$	2) No precipitate is formed but a redox reaction occurs
R) $Na_2S_2O_3 + KI_3(Aq.) \rightarrow$	3) A violet colouration formed as intermediate
S) $Fe^{3+} + S_2O_3^{2-} \rightarrow$	4) neutral gas is evolved

Codes :

(A)

P	Q	R	S
4	1	3	2

(B)

P	Q	R	S
4	1	2	3

(C)

P	Q	R	S
1	2	3	4

(D)

P	Q	R	S
1	4	2	3

Q.37

Which of the following order is correct for increasing p-character in orbital used for bonding by central atom

(A) $SiH_4 > CH_4$

(B) $H_2S > H_2O$

(C) $PH_4^+ > PH_3$

(D) $NH_3 > PH_3$

Q.38

Which one of the following molecular formula cannot be represented by more than two types of functional group isomers?

(A) C_4H_8O

(B) $C_4H_{10}O$

(C) C_3H_9N

(D) $C_4H_8O_2$

Q.39

In acidic medium $KMnO_4$ is decolourise by :-

(a) $H_2C_2O_4$

(b) HNO_3

(c) $Na_2S_2O_3$

(d) HNO_2

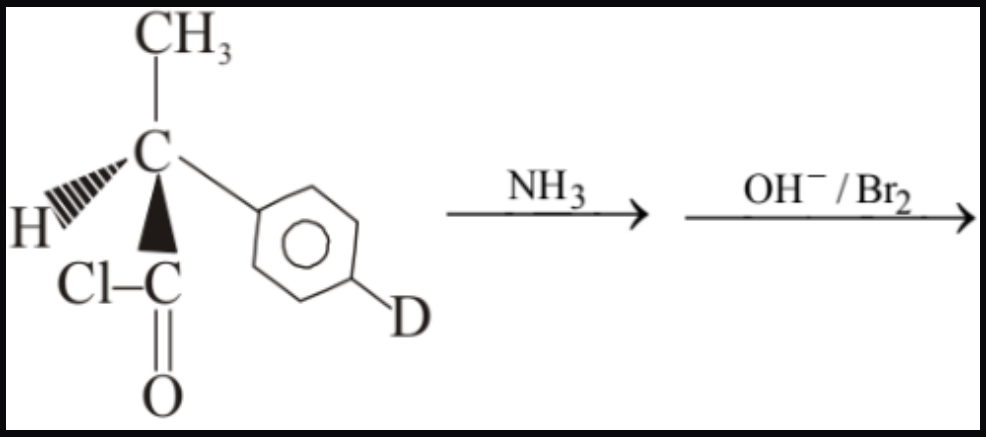
(A) a,b,c,d

(B) a & c only

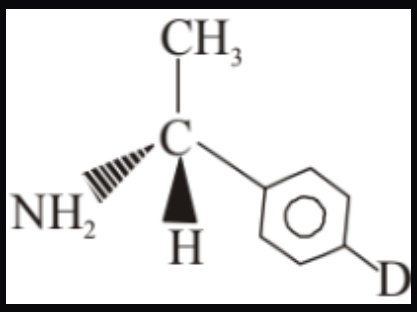
(C) a & d only

(D) c & d only

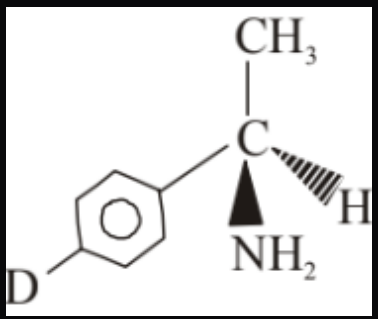
Q.40



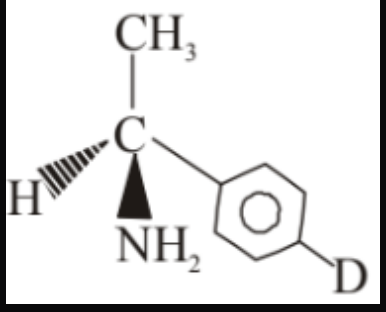
(A)



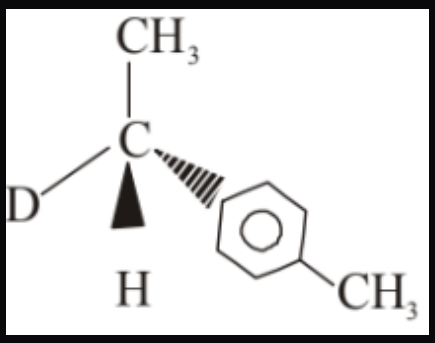
(B)



(C)



(D)



Q.41

One gram of the silver salt of an organic dibasic acid yields, on strong heating, $0.6\ \mathrm{g}$ of silver aproximately. Determine the molecular formula of the acid. [Atomic weight of $\mathrm{Ag} = 108$]

(A) $\mathrm{C_4H_6O_4}$

(B) $\mathrm{C_4H_6O_6}$

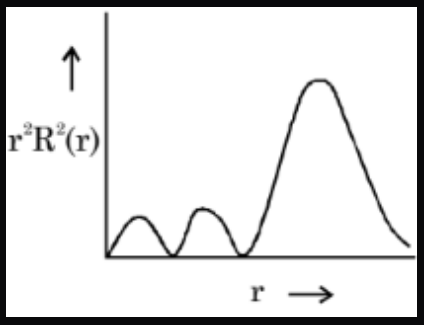
(C) $\mathrm{C_2H_6O_2}$

(D) $\mathrm{C_5H_6O_5}$

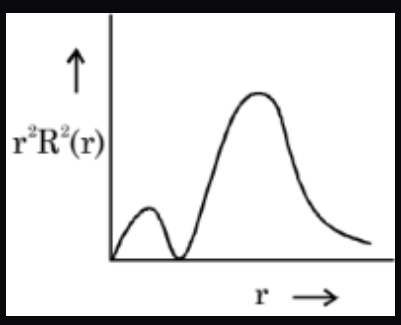
Q.42

Which of the following radial distribution graph corresponds to $l = 2$ for H-atom having least value of ' n ' for which $l = 2$ is allowed

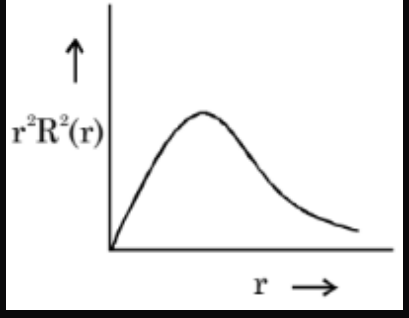
(A)



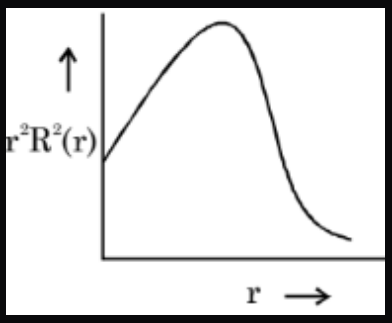
(B)



(C)



(D)

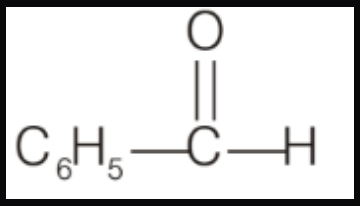


Q.43

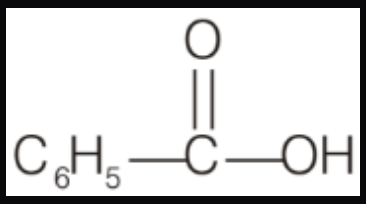
$$\text{Hexane} \xrightarrow[\substack{773K \\ 10\text{-}20\ atm}]{Cr_2O_3} X \xrightarrow[\text{Anhydrous } AlCl_3]{CH_3COCl} Y$$

Y in the above sequence of reaction is

(A)

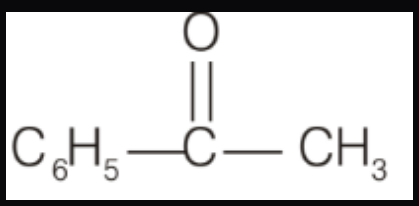


(B)



(C) $C_6H_5 - CH_3$

(D)



Q.44

The decrease stability of higher oxidation state in p-block with increasing atomic number down the group is due to:

(A) decrease in bond energy as going down the group.

(B) energy required to unpair ns^2- electrons is not compensated by the energy released in forming the two additional bonds.

(C) both are correct.

(D) none is correct.

Q.45

Which of the following reaction is used to produce aldehyde?

(A) $\mathrm{CH_3CH_2COOH} \xrightarrow[\text{ii) } \mathrm{H_3O^+}]{\text{i) } \mathrm{LiAlH_4}/\text{ ether}}$

(B) $\mathrm{CH_3COCl} \xrightarrow{\mathrm{R_2CuLi}}$

(C) $\mathrm{CH_3COOH}\ (\text{vapours}) \xrightarrow[\Delta]{\mathrm{MnO}}$

(D) $\mathrm{CH_3CN} \xrightarrow[\text{ii) } \mathrm{H_3O^+}]{\text{i) } \mathrm{SnCl_2/HCl}}$

Q.46

The bond order of O_2^+, O_2, O_2^- and O_2^{2-} will be $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}$ respectively, Find $\mathrm{A} + 2\ \mathrm{B} + \mathrm{C} - \mathrm{D}$.

Q.47

A tetrapeptide has $-\mathrm{COOH}$ group on alanine. The produces glycine (Gly), valine (Val), phenyl alanine (Phe) and alanine (Ala), on complete hydrolysis. For this tetrapeptide, the number of possible sequences (primary structures) with $-\mathrm{NH_2}$ group attached to a chiral centre is

Q.48

If k is rate constant and A is collision frequency then find $(\mathrm{Ea_2} - \mathrm{Ea_1})$ for reaction in $\mathrm{kcal/mol}$ at $227^\circ\mathrm{C}$

$\mathrm{k_1} = 100\mathrm{k_2}$

$10\ \mathrm{A_1} = \mathrm{A_2}$

$\ln 10 = \dfrac{7}{3}$

Q.49

Two mole of an ideal gas is isothermally compressed in two stages starting from (1 atm, 20 lit) to 10 lit. at external pressure $\mathrm{P_1}$ then upto 2 lit at external pressure $\mathrm{P_2}$. Find work done (lit-atm) in the process.

Q.50

11 moles of $\mathrm{FeSO_4}$ are treated with acidic $\mathrm{KMnO_4}$ solution containing its $1\ \mathrm{mole}$ and resulting solution is further

treated with acidic $\mathrm{K_2Cr_2O_7}$ solution as a result remaining $\mathrm{FeSO_4}$ is completely oxidised. Find out minimum number of moles of $\mathrm{K_2Cr_2O_7}$ present in the solution

Q.51

Let $P(x)$ be the polynomial x^3+ax^2+bx+c, where $a, b, c \in R$. If $P(-3)=P(+2)=0$ and $\mathrm{P}'(-3)<0$, which of the following is a possible value of ' c '?

(A) -27

(B) -18

(C) -6

(D) -3

Q.52

If $a_1, a_2, a_3, \ldots a_{4001}$ are terms of an AP such that $\frac{1}{a_1 a_2}+\frac{1}{a_2 a_3}+\ldots+\frac{1}{a_{4000} a_{4001}}=10$ and $\mathrm{a_2+a_{4000}=50}$ then $\left|\mathrm{a_1-a_{4001}}\right|$ is equal to

(A) 20

(B) 30

(C) 40

(D) 10

Q.53

The co-ordinates of the perpendicular drawn from the point $2\hat{i}-\hat{j}+5\hat{k}$ to the line $\vec{r}=(11\hat{i}-2\hat{j}-8\hat{k})+\lambda(10\hat{i}-4\hat{j}-11\hat{k})$ are

(A) $(1,-2,3)$

(B) $(1,2,-3)$

(C) $(-1,2,3)$

(D) $(1,2,3)$

Q.54

Consider a real valued continuous function f such that
$\mathrm{f(x)=\sin x+\int_{-\pi/2}^{\pi/2}(\sin x+tf(t))dt}$
If M and m are maximum and minimum value of the function f, then

(A) $\frac{M}{m}=4$

(B) $\mathrm{M-m=2\pi+1}$

(C) $\mathrm{M+m=4(\pi+1)}$

(D) $\mathrm{Mm=2\left(\pi^2+1\right)}$

Q.55

The line passing through the extremity A of the major axis and extremity B of the minor axis of the ellipse $x^2 + 9y^2 = 9$ meets the auxiliary circle at the point M. The area of the triangle AMO is

(A) $\frac{31}{10}$

(B) $\frac{29}{10}$

(C) $\frac{21}{10}$

(D) $\frac{27}{10}$

Q.56

Let ω be a cube root of unity. If three fair dice are thrown and $\mathbf{r}_1, \mathbf{r}_2$ and $\mathbf{r}_3$ are number obtained on the dice then probability the $\omega^{\mathrm{r}_1} + \omega^{\mathrm{r}_2} + \omega^{\mathrm{r}_3} = 0$, is

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{2}{9}$

(D) $\frac{1}{9}$

Q.57

Area (in square units) bounded by the curves $f(x) = (\cos x + \sec x)^2$ and $g(x) = (\sin x + \operatorname{cosec} x)^2$ between $x = \frac{\pi}{4}$ and $x = \frac{\pi}{3}$ is of the form $\frac{a - b\sqrt{c}}{d\sqrt{c}}$ then ' $a + b + c + d$ ' is

(A) 50

(B) 36

(C) 64

(D) 72

Q.58

If the complex numbers z_1 and z_2 are the solutions of the equation $z^2 - z + 1 = 0$, then the value of the expression $E = \left(z_1^4 - z_1^3 + 2z_1^2 - 2z_1 + 1\right)^{2013} + \left(z_2^4 - z_2^3 + 2z_2^2 - 2z_2 + 1\right)^{2013}$, is

(A) 1

(B) 2

(C) -200

(D) -2

Q.59

The lines $\mathbf{r} = \mathbf{i} - \mathbf{j} + \lambda(2\mathbf{i} + \mathbf{k})$ and $\mathbf{r} = (2\mathbf{i} - \mathbf{j}) + \mu(\mathbf{i} + \mathbf{j} - \mathbf{k})$ intersect for

(A) $\lambda = 1, \mu = 1$

(B) $\lambda = 2, \mu = 3$

(C) all values of λ and μ

(D) no value of λ and μ

Q.60

In an experiment with 15 observations the results were available and $\sum X^2 = 2830, \sum X = 170$, one observation that was 20, was found wrong and was replaced by the correct value 30, then the corrected variance is

(A) 78.00

(B) 188.66

(C) 83.30

(D) 177.33

Q.61

If $f(\mathrm{x})$ is a positive differentiable function satisfying $f(\mathrm{x}+\mathrm{y}+\mathrm{z}) = f(\mathrm{x}) \cdot f(\mathrm{y}) \cdot f(\mathrm{z}) \forall \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathrm{R}$ and $f(2) = 4, f'(0) = 2$, then $f'(2)$ equals

(A) 6

(B) 4

(C) 8

(D) 12

Q.62

If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$, then the value of $x^{100} + y^{100} + z^{100} - \frac{9}{x^{101}+y^{101}+z^{101}}$ is equal to

(A) 0

(B) 3

(C) -3

(D) 9

Q.63

The solution of the differential equation $xy \log\left(\frac{x}{y}\right) dx + \left\{y^2 - x^2 \log\left(\frac{x}{y}\right)\right\} dy = 0$ is

(A) $\frac{x^2}{2y^2} \log\left(\frac{x}{y}\right) - \frac{x^2}{4y^2} = -\log y + \log c$

(B) $\frac{x^2}{2y^2} \log\left(\frac{x}{4y}\right) + \frac{x^2}{4y^2} = \log y + \log c$

(C) $\log\left(\frac{x}{y}\right) - \frac{x^2}{4y^2} + \frac{x^2}{2y^2} \log y + \log c = 0$

(D) none of these

Q.64

Let $f(x)=\cos^2 x; g(x)=\int_0^{1/x} e^{t^2}\,dt$. The value of $\lim_{x\to 0}(f(x))^{g(x)}$ is

(A) 1

(B) e

(C) e^{-1}

(D) 0

Q.65

If x, y are even natural numbers but z is odd natural number, then total possible solutions of the equation $\mathrm{x}+\mathrm{y}+\mathrm{z}=51$ are given by ${}^{(10a+b)}\mathrm{C}_{\mathrm{a}}$, then $(\mathrm{a}+\mathrm{b})$ is equal to

(A) 6

(B) 10

(C) 12

(D) 7

Q.66

The coefficient of x^4 appearing in the expansion of $\left(1+x^2\right)^{10}(1-x)^6\left(1+x^3\right)^8(1+x)^6$ is

(A) -10

(B) 60

(C) 45

(D) 0

Q.67

$S=\{1,2,3\},\ f: S \to S$ satisfies the property : $\forall x \in S,\ f(f(x))=f(x)$. How many different functions are there for $f(x)$?

(A) 8

(B) 10

(C) 1

(D) 4

Q.68

If $\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} = k(\alpha a+\beta b+\gamma c)^3$, then $(2\alpha+\beta-\gamma)^k$ is $(\alpha,\beta,\gamma,k \in z^+)$

(A) 4

(B) 9

(C) 8

(D) 27

Q.69

A rectangle PQRS has sides $\mathrm{PQ}=11$ and $\mathrm{QR}=5$. A triangle ABC has P as orthocentre, Q as circumcentre, R as mid point of BC and S as the foot of altitude from A. Then length of BC is k, where $\mathrm{k}/4$ is equal to

(A) 6

(B) 5

(C) 3

(D) 7

Q.70

If radii of the smallest and largest circle passing through origin and touching the circle $x^2+y^2+4x+6y-3=0$ are r_1 and r_2 respectively, then r_1r_2 is equal to -

(A) $\frac{1}{4}$

(B) $\frac{7}{4}$

(C) $\frac{3}{4}$

(D) $\frac{9}{4}$

Q.71

Given $A=\begin{bmatrix}2 & 0 & -\alpha \\ 5 & \alpha & 0 \\ 0 & \alpha & 3\end{bmatrix}$ For $\alpha \in R-\{a,b\}$, A^{-1} exists and $A^{-1}=A^2-5bA+cI$, when $\alpha=1$. The value of $a+5b+c$ is :

Q.72

Let $f(x)$ and $g(x)$ are differentiable function of x such that
$x^2(gf(x))\cdot(f'g(x))\cdot(g'(x))=\left(1-2x^2\right)(g'f(x))\cdot(f'(x))\,f(g(x))\forall, x\in R.$
Moreover $g(x)$ is positive and $f(x)$ is non negative and $\int_0^k f(g(x))dx=1-e^{-k^2}$, then the value of $\ln\left(\frac{gf(3)}{gf(2)}\right)$ is P, find 4P.

Q.73

Let $f(x)=\lim_{n\to\infty}\tan^{-1}\left\{4n^2\left(1-\cos\frac{x}{n}\right)\right\}$ and $g(x)=\lim_{n\to\infty}\frac{n^2}{2}\ln\cos\left(\frac{2x}{n}\right)$, then the least integer more than $e^{\frac{3}{4}L}$, where $L=\lim_{x\to0}\frac{e^{-2g(x)}-e^{f(x)}}{x^6}$, is

Q.74

Let $f(\mathrm{x})$ be a function such that $\sum_{\mathrm{i}=0}^{9} f(\mathrm{x}+10\mathrm{i})=\sum_{\mathrm{i}=1}^{10} f(\mathrm{x}+10\mathrm{i}-5)$ and if $\int_0^{200} f(100-x)dx=3$. Then the value of $\int_0^{600} f(x)dx$ is

Q.75

If $\overrightarrow{\mathrm{a}}$ and $\overrightarrow{\mathrm{b}}$ are any two unit vectors, then the minimum value of $\frac{1}{|\vec{a}+\vec{b}|^2}+\frac{1}{|\vec{a}-\vec{b}|^2}$ is

Answers & Solutions

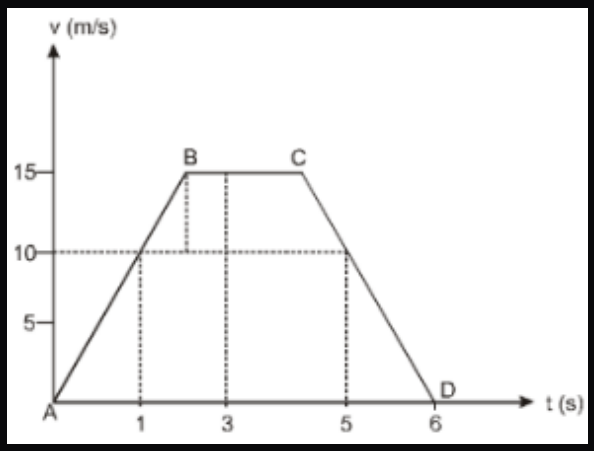
Q.1 Answer:

$1\ \mathrm{N}, 0, -1\ \mathrm{N}$

Solution:

$\mathrm{F}=\ \mathrm{ma}\quad$ [Newtons II law]

$\mathrm{a}=$ slope of velocity time graph



at $t=1$ second, slope of curve is some as slope of line AB.

$$\Rightarrow \mathrm{a}=\frac{10-0}{1-0}=10\ \mathrm{m/s^2}$$

$$\Rightarrow \mathrm{F}=0.1\times 10=1\ \mathrm{N}$$

Similarly at $t=3$ second, slope of curve is same as slope of line BC.

$$\Rightarrow \mathrm{a}=\frac{15-15}{4-2}=0\ \mathrm{m/s^2}$$

$$\Rightarrow \mathrm{F}=0.1\times 0=0\ \mathrm{N}$$

at $t=5$ second, slope of curve is same as that of CD.

$$\Rightarrow \quad \mathrm{a}=\frac{0-10}{6-5}=-10\ \mathrm{m/s^2}$$

$$\mathrm{F}=0.1\times(-10)=-1\ \mathrm{N}$$

Q.2 Answer:

$7\ \mathrm{W}$

Solution:

The energy of incident photons is given by

$\mathrm{h}v = \mathrm{eV_s} + \phi_0 = 2 + 5 = 7\mathrm{eV}$

($\mathrm{V_s}$ is stopping potential and ϕ_0 is work function)

Saturation current $= 10^{-5}\ \mathrm{A} = \frac{\eta \mathrm{P}}{\mathrm{h}\nu}\mathrm{e}$

$= \frac{10^{-5}\mathrm{P}}{7\times \mathrm{e}}\mathrm{e}$ (η is photo emission efficiency)

$\therefore P = 7\ \mathrm{W}.$

Q.3 Answer:

velocity

Solution:

From dimension logic, the quantities of same dimensions can be added or subtracted, so

$[B\mathrm{X}] = [Dt]$ similarly,

$$\Rightarrow \quad \left[\frac{x}{t}\right] = \left[\frac{D}{B}\right] \Rightarrow \left[\frac{D}{B}\right] = \frac{[\mathrm{L}]}{[T}$$

$$\Rightarrow \quad \text{Dimension of } \frac{D}{B} = \left[\mathrm{LT}^{-1}\right]$$

$\left[LT^{-1}\right]$ is the dimension of velocity.

Q.4 Answer:

$\sqrt{2}\mathrm{x}$

Solution:

$$\text{Spacing between successive nodes } = \frac{\lambda}{2}$$

using $\mathrm{V} = \mathrm{n}\lambda$

$$\lambda \propto \mathrm{V} \propto \sqrt{\frac{\mathrm{T}}{\mu}}$$

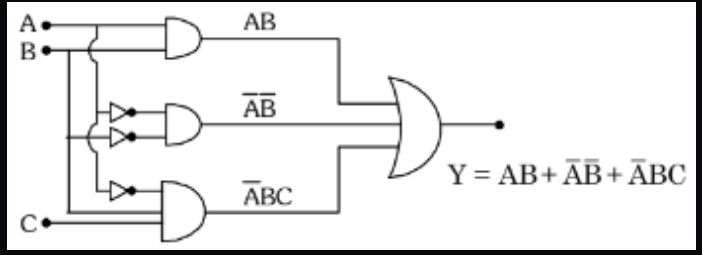
$$\text{New } \frac{\lambda'}{\lambda} = \frac{\sqrt{2\,\mathrm{T}}}{\sqrt{\mathrm{T}}} \Rightarrow \lambda' = \sqrt{2}\lambda$$

$$\frac{\lambda'}{2} = \sqrt{2} \cdot \frac{\lambda}{2} = \sqrt{2}\mathrm{x}$$

Q.5 Answer:

For $\mathrm{A} = 0,\ \mathrm{B} = 1, \mathrm{C} = 1$; Output $= 1$

Solution:



Q.6 Answer:

$2\ \Omega$

Solution:

(Initial part)

$\frac{x}{100}-x=\frac{2}{3} \Rightarrow x=40\ \mathrm{cm}$

If there is shifting by $22.5\ \mathrm{cm}$. Then, to obtain the balance point in meter bridge

$$\frac{2(3+x)}{3x}=\frac{62.5}{37.5} \Rightarrow (6+2x)37.5=62.5\times 3x$$

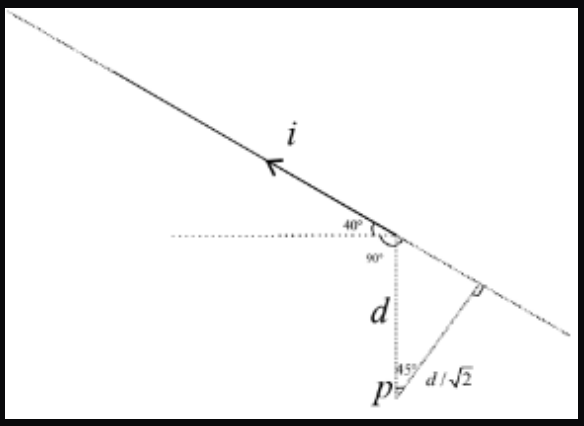
$$\Rightarrow \quad 225+75x=187.5x$$

$$187.5x-75x=225 \Rightarrow 112.5x=225 \Rightarrow x=2$$

Q.7 Answer:

$\frac{\mu_0 i}{4\pi d}(\sqrt{2}-1)$

Solution:



$B=\frac{\mu_0 i}{4\pi d}\times[\sin\theta_1+\sin\theta_2]$

Where d is $\perp$ distance iron wire

$$B=\frac{\mu_0 i}{4\pi d/\sqrt{2}}\times[\sin 90-\sin 45^\circ]$$

$$=\frac{\sqrt{2}\mu_o i}{4\pi d}\left[1-\frac{1}{\sqrt{2}}\right] \quad B=\frac{\mu_0 i}{4\pi d}[\sqrt{2}-1]$$

Q.8 Answer:

$\frac{1}{4\pi\varepsilon_0}\frac{2\mathrm{q}}{\mathrm{L}}\left(1-\frac{1}{\sqrt{5}}\right)$

Solution:

By the principle of superposition, we have

$$\mathrm{V}=\sum \frac{1}{4\pi\varepsilon_0}\cdot\frac{\mathrm{q}}{\mathrm{r}}$$

$$\mathrm{V(A)}=\frac{2}{4\pi\varepsilon_0}\cdot\frac{\mathrm{q}}{\mathrm{L}}-\frac{2}{4\pi\varepsilon_0}\cdot\frac{\mathrm{q}}{\sqrt{5}\,\mathrm{L}}$$

$$\mathrm{V(\ A)}=\frac{2\mathrm{q}}{4\pi\varepsilon_0\,\mathrm{L}}\left[1-\frac{1}{\sqrt{5}}\right]$$

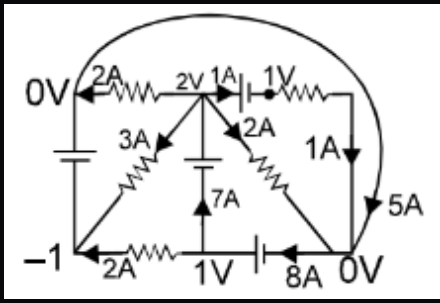
Q.9 Answer:

$1\ \text{A}, 22\ \text{W}$

Solution:

(1) Reading of ammeter $= 1\text{amp}$

(2) Thermal energy, $\text{P} = \text{I}^2\text{R}$



$$\begin{aligned}\text{P} &= 3^2 \times 1 + 2^2 \times 1 + 2^2 \times 1 + 2^2 \times 1 + 1^2 \times 1 \\ &= 9 + 4 + 4 + 4 + 1 \\ &= 22\ \text{Watt.}\end{aligned}$$

Q.10 Answer:

$R/3$

Solution:

$\text{v} = \frac{50}{100}\ \text{V}_\text{e} = \frac{1}{2}\sqrt{\frac{2\text{GM}}{\text{R}}}$

Applying energy conservation

$$\Rightarrow -\frac{GMm}{R} + \frac{1}{2}mv^2 = -\frac{GMm}{(R+h)}$$

$$v^2 = \frac{2GM}{R} - \frac{2GM}{R+h}$$

$$\Rightarrow \quad \frac{1}{4} \cdot \frac{2GM}{R} = 2GM\left(\frac{1}{R} - \frac{1}{R+h}\right)$$

$$\Rightarrow \quad \frac{1}{4R} = \frac{h}{R(R+h)} \Rightarrow R + h = 4h \Rightarrow h = R/3$$

Q.11 Answer:

$R = 8\varepsilon_0 \frac{T}{\sigma^2}$

Solution:

The pressure due to surface tension $= \frac{4T}{R}$

The pressure due to electrostatic forces $= \frac{\sigma^2}{2\varepsilon_0}$

Just before the bubble bursts.

$\frac{4T}{R} = \frac{\sigma^2}{2\varepsilon_0} \quad \text{or} \quad R = \frac{8T\varepsilon_0}{\sigma^2}$

Q.12 Answer:

4.405 mm

Solution:

$\mathrm{L.C} = \frac{0.5}{100}\ \mathrm{mm} = 0.005\ \mathrm{mm}$
Zero error $= 2 \times 0.005\ \mathrm{mm} = 0.01\ \mathrm{mm}$
So zero correction is $= -0.01\ \mathrm{mm}$
M.S.R = $4\ \mathrm{mm}$
C.S.R $= 83 \times L.C = 0.415\ \mathrm{mm}$
So diameter $= 4\ \mathrm{mm} + 0.415\ \mathrm{mm} - 0.01\ \mathrm{nm} = 4.405\ \mathrm{mm}$

Q.13 Answer:

4.8 V

Solution:

Given, $\mathrm{R} = 3\Omega, \mathrm{X_L} = 4\Omega$
In $\mathrm{L-R}$ **circuit,** $\tan\phi = \frac{\mathrm{X_L}}{\mathrm{R}} = \frac{4}{3} \Rightarrow \phi = 53^\circ$ **Current across inductor**
$I = I_0 \sin\left(wt - 53^0\right) = \frac{v_0}{z} \sin\left(wt - 53^0\right)$
$= \frac{10}{\sqrt{3^2+4^2}} \sin\left(\mathrm{wt} - 53^0\right) = 2\sin\left(w\mathrm{wt} - 53^0\right)$
$\therefore$ **voltage across the inducter at** $\mathrm{t} = \mathrm{T}/2$
$(\mathrm{V_L})_0 = \mathrm{I_0 X_L} = 2 \times 4 = 8$
$\mathrm{V_L} = (\mathrm{V_L})_0 \sin\left(\mathrm{wT}/2 - 53^0 + \pi/2\right)$
$= 8\sin\left(\pi - 53^0 + \pi/2\right)$
$|\mathrm{V_L}| = 8\sin 37^0 = 8 \times \dfrac{3}{5} = 4.8\ \mathrm{V}$

Q.14 Answer:

600 W

Solution:

Speed of each bullet, $v = 360\ \mathrm{km/h}$
$= 360 \times \frac{5}{18}\ \mathrm{m/s} = 100\ \mathrm{m/s}$
Number of bullets fired per minute, $N = 360$
Kinetic energy of 360 bullets,
k' = $N\times$ **kinetic energy of each bullet**
$= 360 \times \dfrac{1}{2}mv^2 = 360 \times \dfrac{1}{2} \times 0.02 \times 100^2$
$= 3.6 \times 10^4\ \mathrm{J}$
$\therefore\ \text{Power of gun} = \dfrac{\text{Total energy}}{\text{Time}}$
$= \dfrac{3.6 \times 10^4}{60} = 600\ \mathrm{W}$
[$\because$ **Time, 1 minute** $= 60\ \mathrm{s}$]

Q.15 Answer:

$-\frac{50^\circ}{9}$ per mm

Solution:

$\theta = a + \frac{b}{\lambda^2}$

$30 = a + \frac{b}{(5000)^2}$ **and** $50 = a + \frac{b}{(4000)^2}$

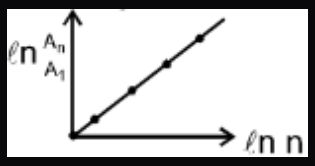
Solving for a, we get $a = -\frac{50^\circ}{9}$ **per** mm

Q.16 Answer:

will have certain points lying on a straight line with slope 4

Solution:

$$A_n = \pi r^2 = \pi \left(r_0 n^2\right)^2$$
$$= \pi r_0^2 n^4$$

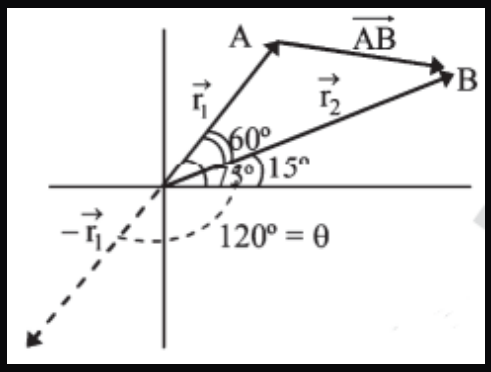


$$\ell \mathrm{n} \frac{\mathrm{A_n}}{\mathrm{A_1}} = \ell \mathrm{n}\left(\mathrm{n}^4\right) = 4 \ell \mathrm{nn}$$

Q.17 Answer:

9

Solution:



$\because$ **Displacement** $|\overrightarrow{\mathrm{AB}}| = |\vec{r}_2 - \vec{r}_1|$

& **angle between** $\vec{r}_2$ & $-\vec{r}_1 = 120^\circ$

$\therefore$ **According to law of parallelogram**

$$\left|\overrightarrow{r_2} - \vec{r}_1\right| = \sqrt{r_2^2 + r_1^2 + 2 r_1 r_2 \cos\theta}$$
$$= \sqrt{4^2 + 3^2 + 2 \times 4 \times 3 \times \cos 120^\circ}$$
$$= \sqrt{16 + 9 + 24(-1/2)} = \sqrt{13}$$

Q.18 Answer:

Maximum acceleration of $4\ \mathrm{kg}$ block is $2\ \mathrm{m/s^2}$

Solution:

So block ' Q **' is moving due to force while block 'P' due to friction.**

Friction direction on both $+\mathrm{Q}$ **blocks as shown.**

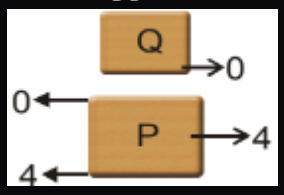




First block ' Q ' will move and P will move with ' Q ' so by FBD taking ' P ' and ' Q ' as system

$$\mathrm{F}-9=0 \quad \Rightarrow \mathrm{F}=9 \mathrm{~N}$$

When applied force is $4 \mathrm{~N}$ then FBD



$4 \mathrm{~kg}$ block is moving due to friction and maximum friction force is $8 \mathrm{~N}$.

So acceleration $=\frac{8}{4}=2 \mathrm{~m/s^2}=\mathrm{a_{max}}$.

Slipping will start at when Q has +ve acceleration equal to maximum acceleration of P i.e. $2 \mathrm{~m/s^2}$.

$$\mathrm{F}-17=5 \times 2 \quad \Rightarrow \mathrm{F}=27 \mathrm{~N}.$$

Q.19 Answer:

They have a certain constant average speed

Solution:

$$\mathrm{V_{av}}=\sqrt{\frac{8 \mathrm{KT}}{\pi \mathrm{m}}}, \text{ as } \mathrm{T}= \text{ constant}$$

$$\therefore \quad \mathrm{V_{av}}= \text{ constant}$$

Q.20 Answer:

$\mathrm{A} \rightarrow \mathrm{R}, \mathrm{B} \rightarrow \mathrm{Q}, \mathrm{C} \rightarrow \mathrm{S}, \mathrm{D} \rightarrow \mathrm{P}$

Solution:

(A) X-rays $\rightarrow$ (R) Crystal structure study

(B) uv- rays $\rightarrow$ (Q) Finger prints in forensic labs

(C) Radio waves (S) TV communication system

(D) IR-rays (P) Remote switches

Q.21 Answer:

—

Solution:

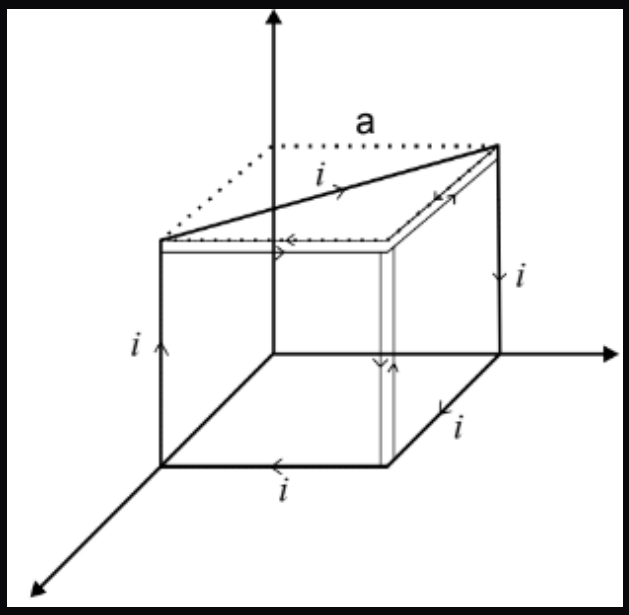
$$U_i = \frac{2kQq}{a} + \frac{kqq}{2a}$$

$$U_f = kQq\left(\frac{1}{a+x} + \frac{1}{a-x}\right) + \frac{kq^2}{2a}$$

Hence, $k = \frac{1}{4\pi\varepsilon_0}$

$|\Delta U| = |U_f - U_i| = \frac{2kQqx^2}{a^3}$ (for $x << a$)

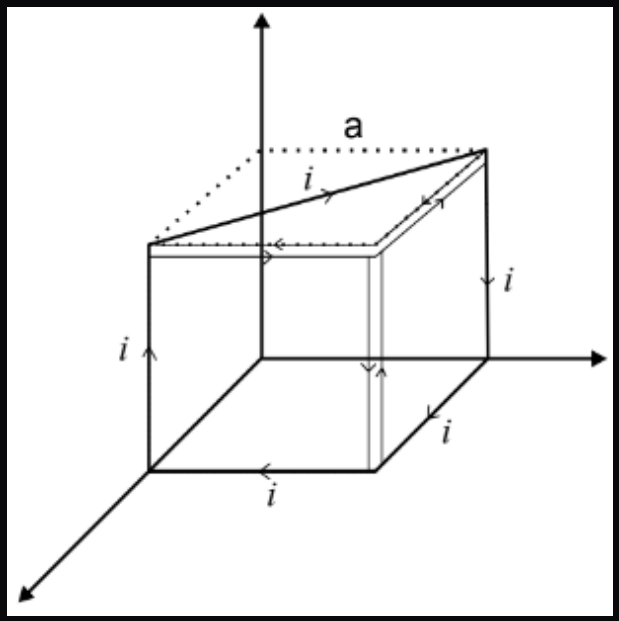
$\therefore \Delta U \alpha x^2$



Q.22 Answer:

—

Solution:



Here $\vec{M} = -ia^2\hat{i} - i\frac{a^2}{2}\hat{j} - ia^2\hat{k}$

$\vec{M} = -ia^2\left(\hat{i} + \frac{\hat{j}}{2} + \hat{k}\right) \Rightarrow |\vec{M}| = \frac{3}{2}ia^2$

$= \frac{3}{2} \times 3 \times 1 \quad |\vec{M}| = \frac{9}{2}$ amphere - meter ^{2}

Q.23 Answer:

—

Solution:

induced emf
$= \mathrm{e} = \mathrm{B_H lv} = 5 \times 10^{-5} \times 3 \times 3$
$= 0.45 \times 10^{-3} = 0.45 \mathrm{mV}$

Q.24 Answer:

—

Solution:

$m_1 = 1$ kg, **extension** $l_1 = 5$ cm $= 5 \times 10^{-2}$ m
$\therefore \quad m_1 g = k l_1$
$k =$ **force constant of the spring**
$k = \frac{m_1 g}{l_1} = \frac{1 \times 10}{5 \times 10^{-2}} = 200$ N/m
Time period of the block of mass 2 kg,
$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{2}{200}}$
$= 2\pi \times \frac{1}{10} = \frac{\pi}{5}$ s
Maximum velocity, $v_{\max} = A\omega$
where, $\quad A =$ **amplitude**
$= 10$ cm $= 10 \times 10^{-2}$ m
$v_{\max} = A \times \frac{2\pi}{T} = 10 \times 10^{-2} \times \frac{2\pi}{\pi/5}$
$= 10^{-1} \times 2 \times 5$
$= 1$ m/s

Q.25 Answer:

—

Solution:

$\angle ABC = \pi - 2(\hat{i} - \hat{r})$

$2\angle ABC = \angle AOC[\text{ half angle theorem}]$

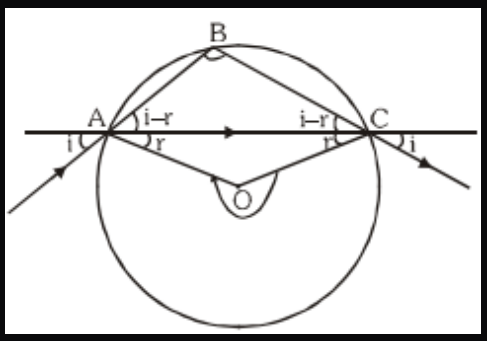
$2\pi - 4\hat{i} + 4r = (\pi + 2r)$

$2\hat{i} - r = \frac{\pi}{z} \Rightarrow r = 2\hat{i} - \frac{\pi}{2}$

$1 \sin \hat{i} = \mu \sin \hat{r}$

$\sin \hat{i} = \mu \sin\left(2\hat{i} - \frac{\pi}{2}\right)$

$\sin \hat{i} = \frac{\sqrt{3}}{2} \Rightarrow \hat{i} = \frac{\pi}{3} \Rightarrow \frac{2\pi}{6} \Rightarrow k = 6$



Q.26 Answer:

Boiling point

Solution:

Separation of liquids into their pure form to from its liquid mixture is done by tractional distillation technique. It is done, when two liquids have difference in their boiling points $(10 - 40\ \mathrm{K})$.

Q.27 Answer:

32

Solution:

$\mathrm{P_{Total}} = \mathrm{P_{HNO_3}} + \mathrm{P_{NO_2}} + \mathrm{P_{H_2O}} + \mathrm{P_{O_2}}$

$\because \mathrm{P_{NO_2}} = 4\mathrm{P_{O_2}} \& \mathrm{P_{H_2O}} = 2\mathrm{P_{O_2}}$

$\therefore \mathrm{P_{Total}} = \mathrm{P_{HNO_3}} + 7\mathrm{P_{O_2}}$

$\Rightarrow 30 - 2 = \mathrm{P_{O_2}} \times 7 \Rightarrow \mathrm{P_{O_2}} = 4$

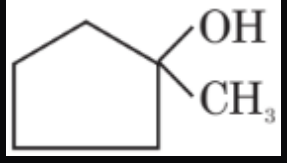
$\mathrm{K_P} = \frac{(\mathrm{P_{NO_2}})^4 \cdot \mathrm{P_{H_2O}} \cdot \mathrm{P_{O_2}}}{\mathrm{P^4HNO_3}}$

$= \frac{(4 \times 4)^4 \times (2 \times 4)^2 \times 4}{2^4} = 2^{20}$

$\mathrm{K_P} = \mathrm{K_C}(\mathrm{RT})^{\Delta\, \mathrm{ng}} = \mathrm{K_C} \times (0.08 \times 400)^3$

$\mathrm{K_C} = \frac{2^{20}}{(32)^3} = 32$

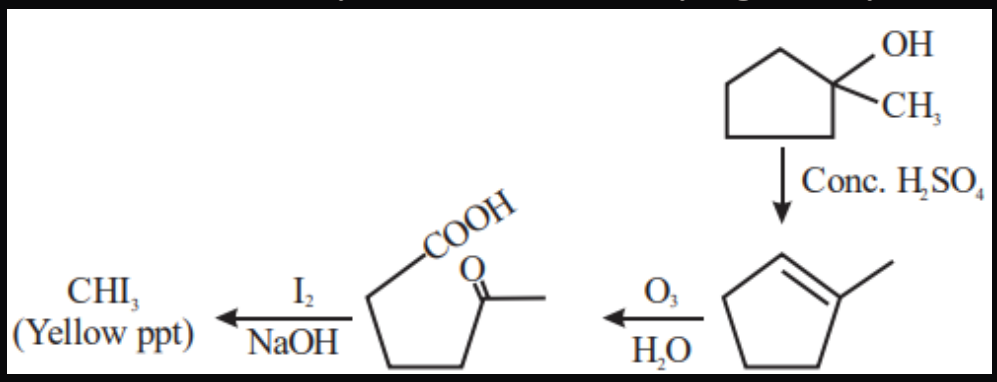
Q.28 Answer:



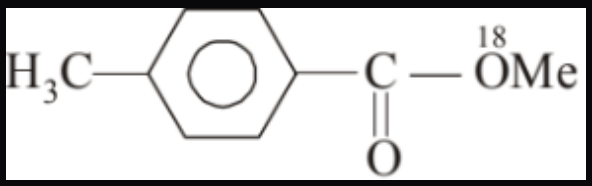
Solution:

A does not change colour with $K_2Cr_2O_7$ i.e. it is a 3° alcohol.

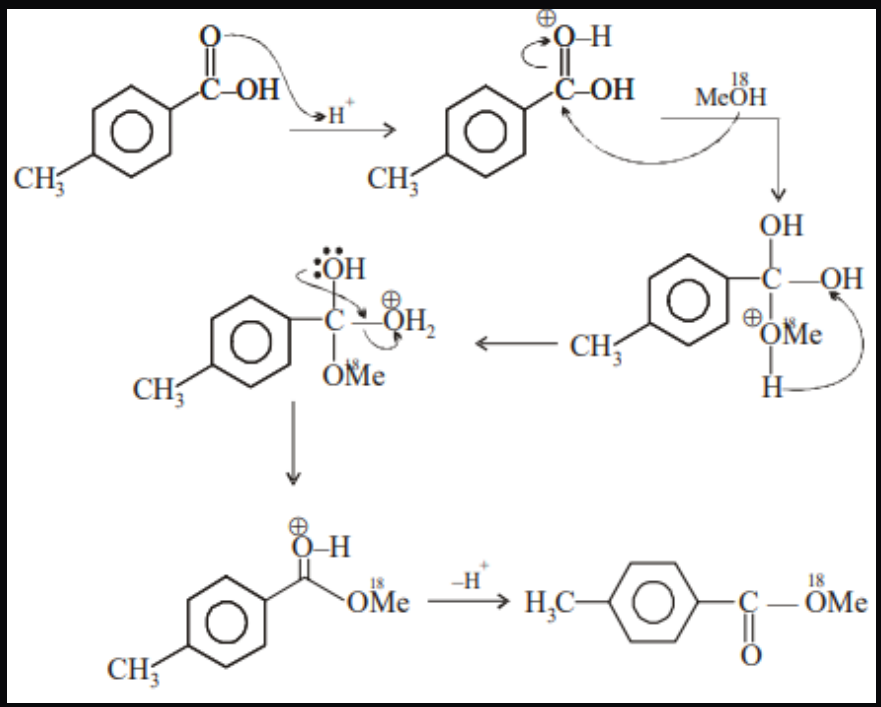
Alkene formed after dehydration oxidative ozonolysis give methyl ketone.



Q.29 Answer:



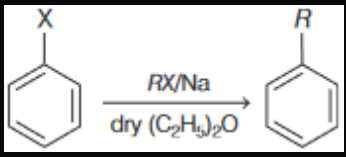
Solution:



Q.30 Answer:

$RX/\text{Na}/\text{dry}\ (C_2H_5)_2\ O$

Solution:



This reaction is known as Wurtz-Fittig reaction where the alkyl halide reacts with aryl halide with sodium in pressure of dry ether.

Q.31 Answer:

3, 3, 3, 1

Solution:

Stability $Pb^{+2} > Sn^{+2} > Ge^{+2}$

in PbI^4, Pb^{+4} is less stable

Acidic nature :

$CO_2 > SiO_2 > GeO_2 > SnO_2 > PbO_2$

$GeO_2SnO_2PbO_2$ = amphoteric

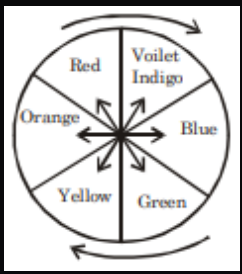
B.E. $= N - 4 > B - 4 > As - 4$

Q.32 Answer:

Absorbed light	Observe colour
Blue	Green

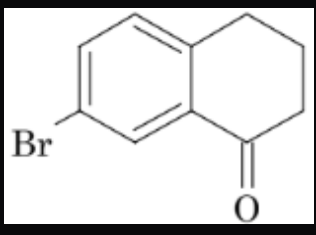
Solution:

According to the complementry colour cycle

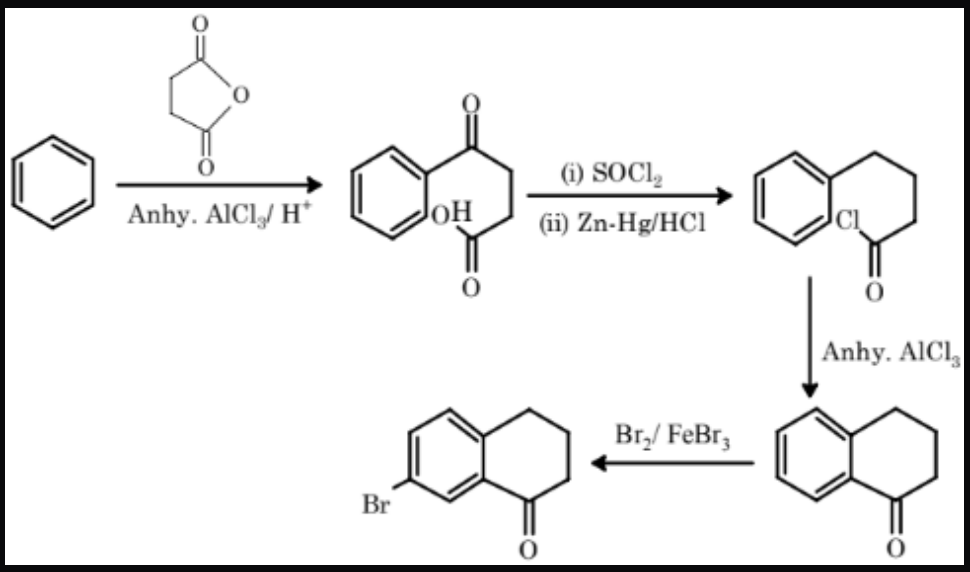


Absorbed		Observed
(1) Yellow	→	Violet
(2) Orange	→	Blue
(3) Green	→	Red
(4) Blue	→	Orange

Q.33 Answer:



Solution:



Q.34 Answer:

Sn, Pb

Solution:

Metals $= \mathrm{Sn}, \mathrm{Pb}$; **Non-metals** $= \mathrm{C}, \mathrm{Si}$; **Metalloids** $= \mathrm{Ge}$.

Q.35 Answer:

ClO_4^-

Solution:

Disproportionation is a specific type of redox reaction in which a species is simultaneously reduced and oxidised to form two different products.

Electronic configuration of Cl **is**

$1s^2 2s^2 2p^6 3s^2 3p^5$

It can exhibit an maximum oxidation state of +7.

In the given question, All species can exhibit disproportionation reaction which simultaneously oxidise as well as reduce.

In ClO_4^- **this cannot undergo oxidation as +7 is maximum oxidation state exhibited by chlorine. Hence, the correct option is (d).**

Q.36 Answer:

P	Q	R	S
1	4	2	3

Solution:

(A) $[\mathrm{Na_2Zn(OH)_4}] + \mathrm{H_2\ S} \rightarrow \mathrm{ZnS} \downarrow + 2\mathrm{NaOH} + 2\mathrm{H_2O}$

(B)

$3\mathrm{Hg} \downarrow + 3\mathrm{HgS} \downarrow + 12\mathrm{HCl} + 4\mathrm{HNO_3} \rightarrow 6\mathrm{HgCl_2} + 3\ \mathrm{S} \downarrow + 4\mathrm{NO} \uparrow + 8\mathrm{H_2O}$

$\mathrm{S} \downarrow + 6\mathrm{HCl} + 2\mathrm{HNO_3} \rightarrow \mathrm{SO_4^{2-}} + 6\mathrm{Cl^{-1}} + 8\mathrm{H^+} + 2\mathrm{NO} \uparrow$

(D)

$Fe^{3+} + 2\,S_2O_3^{2-} \rightarrow [Fe(S_2O_3)_2]^{-1}$ (violet colouration)

$[Fe(S_2O_3)_2]^{-1} + Fe^{3+} \rightarrow 2Fe^{2+} + S_4O_6^{2-}$

Q.37 Answer:

$H_2S > H_2O$

Solution:

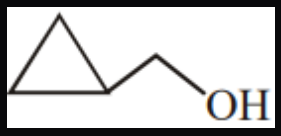
According to drago's rule, in H_2S, bond are formed by p-orbitals 100% p-character while in H_2O bonds are formed by sp^3 (75 % p-character) hybrid orbitals.

Q.38 Answer:

$C_4H_{10}O$

Solution:

(A) $CH_3CH_2COCH_3$,
Ketone ; $CH_3CH_2CH_2CHO$,
Aldehyde;



, Alcohol
(B) $CH_3CH_2CH_2CH_2OH$,
Alcohol ; $CH_3CH_2OCH_2CH_3$, **Ether**
(C) $CH_3CH_2CH_2NH_2$,
1° **amine ;** $CH_3CH_2NHCH_3$, 2° **amine ;** $(CH_3)_3\ N$, 3° **amine**
(D) $CH_3CH_2CH_2COOH$, **Carboxylic acid;** $CH_3CH_2COOCH_3$, **Ester**
$CH_3COCH_2CH_2OH$, **Hydroxy ketone**

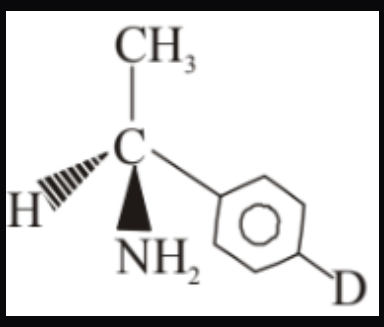
Q.39 Answer:

a & d only

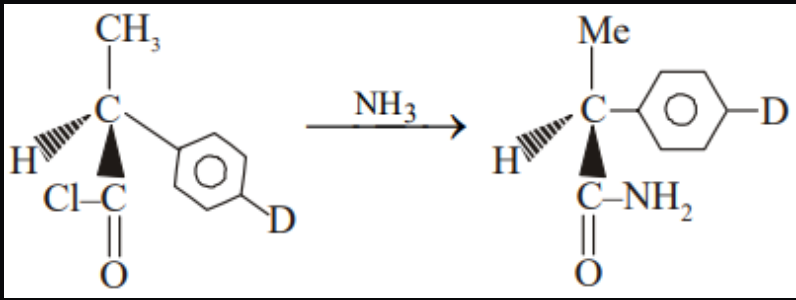
Solution:

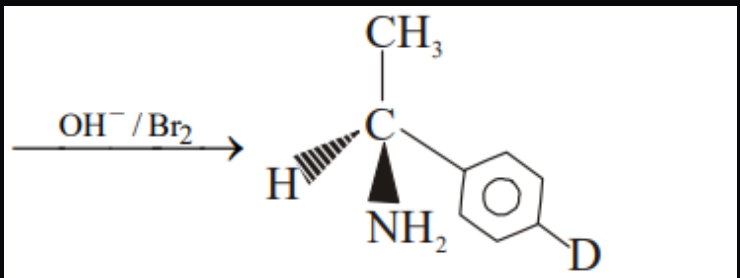
In acidic medium $KMnO_4$ undergoes redox reaction with $H_2C_2O_4$, $Na_2S_2O_3$ and HNO_2 and converted into Mn^{2+}.

Q.40 Answer:



Solution:





(Hoffmann Bromamide)

Q.41 Answer:

$C_5H_6O_5$

Solution:

$$RH_2 \longrightarrow Ag_2R \xrightarrow{\Delta} Ag$$

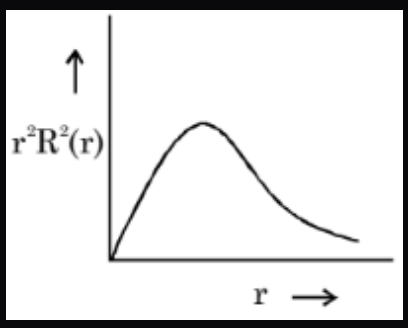
DOAC on Ag

$$2 \times \frac{1}{M_{selt}} = \frac{0.6}{108} \times 1$$

$$M_{Salt} = \frac{2 \times 108}{0.6} = 360$$

wt. of Acid $= 360 - 2 \times 108 + 2 = 146$

Q.42 Answer:



Solution:

In Graph (I) - two radial nodes are present

$\therefore n - l - 1 = 2 \quad n = 5$ for $l = 2$

In Graph (II) one radial node is present

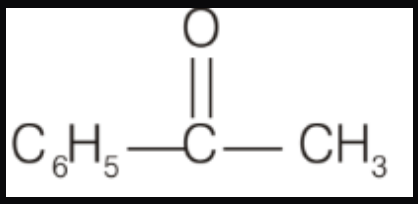
$\therefore n - l - 1 = 2 \quad n = 4$ for $l = 2$

In Graph (III) zero radial node

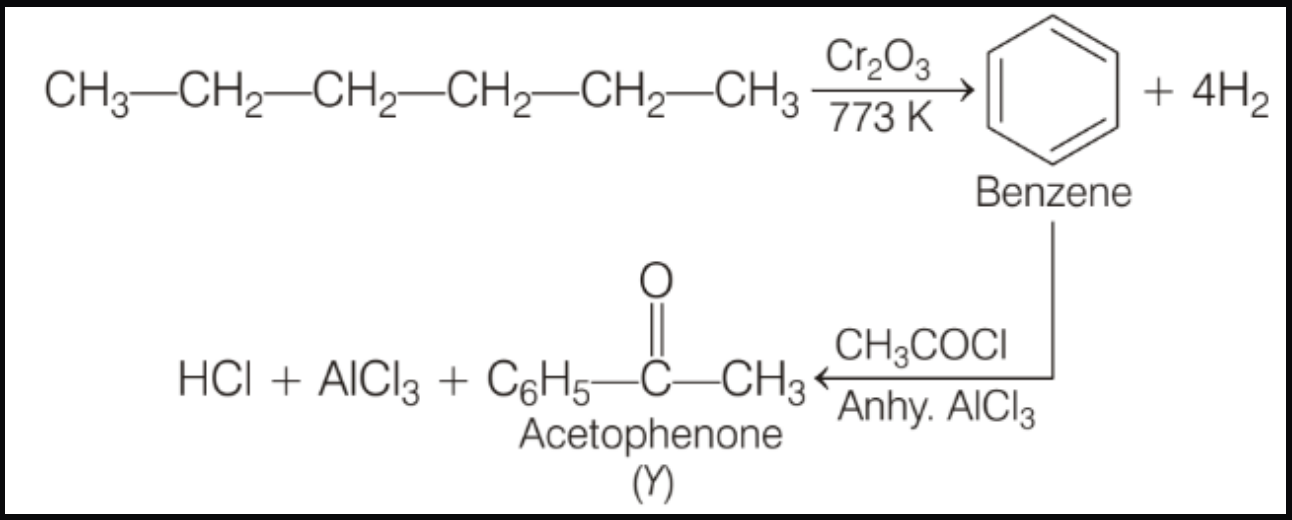
$\therefore n - l - 1 = 2 \quad n = 3$ **for** $l = 2$

Graph (IV) does not represents any radial distribution curve

Q.43 Answer:



Solution:



Q.44 Answer:

both are correct.

Solution:

Size of cation increases down the group and therefore bond energy decreases. Hence the energy available is not sufficient to unpair the electrons for the formation of additional bonds.

Q.45 Answer:

$\mathrm{CH_3CN} \xrightarrow[\text{ii) } \mathrm{H_3O^+}]{\text{i) } \mathrm{SnCl_2/HCl}}$

Solution:

(A) $\mathrm{CH_3CH_2COOH} \xrightarrow[\text{ii) } \mathrm{H_4O^+}]{\text{i) } \mathrm{LiAlI_4/\ ether}} \mathrm{CH_3CH_2CH_2OH}$

(B) $\mathrm{CH_3COCl} \xrightarrow{\mathrm{R_2CuLi}} \mathrm{CH_3COR}$

(C) $\mathrm{CH_3COOH}$ **(vapours)** $\xrightarrow[\Delta]{\mathrm{MnO}} \mathrm{CH_3COCH_3}$

(D) $\mathrm{CH_3CN} \xrightarrow[\mathrm{H_3O^+}]{\mathrm{SnCl_2/HCl}} \mathrm{CH_3CHO}$

Q.46 Answer:

—

Solution:

According to MOT, for the number of electrons > 14, the electronic configuration is -

For

$O_2 - {\sigma_{g^2}}^{(1s)} < {\sigma^{*2}}_{u}^{(1s)} < {\sigma^2}_{g}(2\,\mathrm{s}) < {\sigma^{*2}}_{u}(2\,\mathrm{s}) < {\sigma^2}_{g}(2\mathrm{pz}) < {\pi^2}_{\mathrm{u}}(2\mathrm{px}) = {\pi^2}_{\mathrm{u}}(2\mathrm{py}) < {\pi^{*1}}_{\mathrm{g}}^{(2\mathrm{px})} = {\pi^{*1}}_{\mathrm{g}}^{(2\mathrm{py})} < {\sigma^*}_{\mathrm{u}}^{(2\mathrm{pz})}$

- Number of bonding electrons - 10

- Number of antibonding electrons - 6

- Bond order of $O_2 = \frac{10-6}{2} = 2$

For

- Number of bonding electrons - 10

- Number of antibonding electrons - 7

- Bond order of ${O_2}^- = \frac{10-7}{2} \mathbf{1.5}$

For

${O_2}^+ - {\sigma^2}_{g}(1\,\mathrm{s}) < {\sigma^{*2}}_{u}(1\,\mathrm{s}) < {\sigma^2}_{\mathrm{g}}(2\,\mathrm{s}) < {\sigma^{*2}}_{\mathrm{u}}^{(2\,\mathrm{s})} < {\sigma^2}_{\mathrm{g}}^{(2\mathrm{pz})} < {\pi^2}_{\mathrm{u}}(2\mathrm{px}) = {\pi^2}_{\mathrm{u}}(2\mathrm{py}) < {\pi^{*1}}_{\mathrm{g}}(2\mathrm{px}) = {\pi^*}_{\mathrm{g}}^{(2\mathrm{py})} < {\sigma^*}_{\mathrm{u}}^{(2\mathrm{pz})}$

- Number of bonding electrons - 10

- Number of antibonding electrons - 5

- Bond order of ${O_2}^+ = \frac{10-5}{2} = \mathbf{2.5}$

For

${O_2}^{2-} - {\sigma^2}_{g}(1\,\mathrm{s}) < {\sigma^{*2}}_{\mathrm{u}}^{(1\,\mathrm{s})} < {\sigma^2}_{\mathrm{g}}^{(2\,\mathrm{s})} < {\sigma^{*2}}_{\mathrm{u}}^{(2\,\mathrm{s})} < {\sigma^2}_{\mathrm{g}}^{(2\mathrm{pz})} < {\pi^2}_{\mathrm{u}}^{(2\mathrm{px})} = {\pi^2}_{\mathrm{u}}(2\mathrm{py}) < {\pi^{*2}}_{\mathrm{g}}(2\mathrm{px}) = {\pi^{*2}}_{\mathrm{g}}(2\mathrm{py}) < {\sigma^*}_{\mathrm{u}}(2\mathrm{pz})$

- Number of bonding electrons - 10

- Number of antibonding electrons - 5

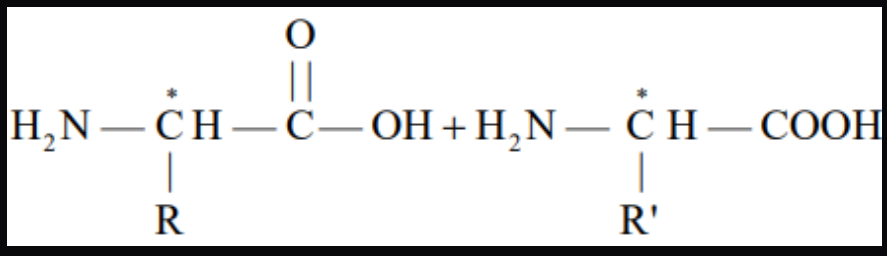
- Bond order of ${O_2}^{2-} = \frac{10-8}{2} = 1$

Therefore, the bond order of ${O_2}^+ = 2.5, O_2 = 2, O_2^- = 1.5, {O_2}^{2-} = 1$

Q.47 Answer:

—

Solution:



C^* is chiral carbon tetrapeptide has four amino acids joined by three peptide linkage.

$-COOH$ group is on alanine part, thus it is at fixed C-terminal position in each combination.

Glycine is optically inactive thus it cannot be on the $N-$ terminal side. Thus possible combinations are

Phe-Gly-Val-Ala, Phe-Val-Gly-Ala,

Val-Gly-Phe-Ala, Val-Phe-Gly-Ala

Q.48 Answer:

—

Solution:

$$\frac{k_1}{k_2} = \frac{A_1}{A_2} e \left(\frac{Ea_2 - Ea_1}{1000} \right)$$

$$100 = \frac{1}{10} e^{\left(\frac{Ba_2 - Ba_2}{1000} \right)}$$

$$1000 = e^{\left(\frac{Ea_2 - Ea_{a_1}}{1000} \right)}$$

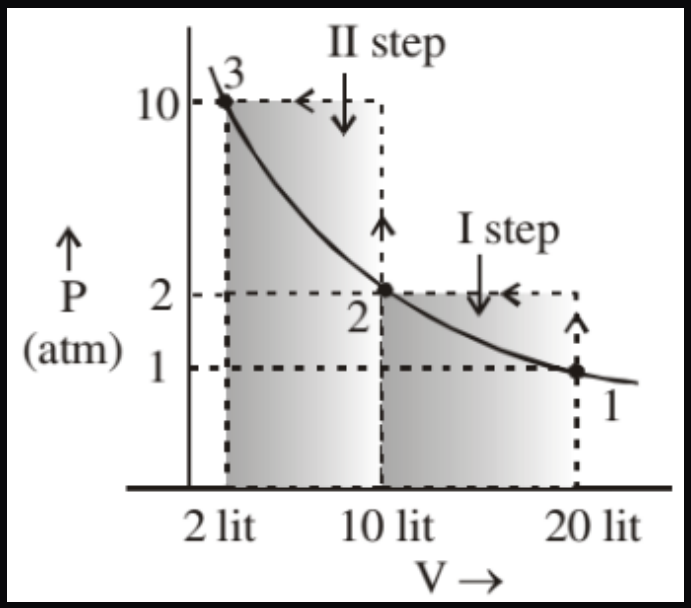
$$\ln 10^3 = \left(\frac{Ba_i - Ea_1}{1000} \right)$$

$$\frac{3 \ln 10 \times 1000}{1000} = (Ea_2 - Ea_1)$$

Q.49 Answer:

—

Solution:



If vol. is 10 lit

$P_2 = \frac{1 \times 20}{10} = 2 \text{ atm}$

If $V = 2$ **lit**

then $P_3 = \frac{2 \times 10}{2} = 10 \text{ atm}$

$$W = (20 - 10) \times 2 + (10 - 2) \times 10$$

$$= 100 \text{ lit } - \text{atm}$$

Q.50 Answer:

—

Solution:

$$\underset{\substack{11 \text{ moles} \\ (\text{Excess} \\ \text{reagent})}}{\mathrm{Fe}^{2+}} + \underset{\substack{1 \text{ mole} \\ (\text{Limiting} \\ \text{reagent})}}{\mathrm{KMnO_4}} \xrightarrow{\mathrm{H}^+} \mathrm{Mn}^{2+} + \mathrm{Fe}^{3+}$$

$$\mathrm{n_{Fe^{2+}}(\text{ reacted }) \times 1 = n_{KMnO_4} \times 5}$$

$$\Rightarrow \mathrm{n_{Fe^{2+}}(\text{ reacted }) = 1 \times 5 = 5 \text{ mole}}$$

So, $\mathrm{n_{Fe^{2+}}(\text{left}) = 11 - 5 = 6}$

Now, reaction with acidic $\mathrm{K_2Cr_2O_7}$

$$\underset{\text{x mole}}{\mathrm{Cr_2O_7^{2-}}} + \underset{6 \text{ mole}}{\mathrm{Fe}^{2+}} \xrightarrow{\mathrm{H}^+} \mathrm{Cr}^{3+} + \mathrm{Fe}^{3+}$$

$$\mathrm{x \times 6 = 6 \times 1 \Rightarrow x = 1}$$

Q.51 Answer:

-27

Solution:

$$P(x) = x^3 + ax^2 + bx + c$$

$$P'(x) = 3x^2 + 2ax + b$$

$$P(-3) = 0$$

$$\Rightarrow -27 + 9a - 3b + c = 0 \quad \ldots(1)$$

$$P(2) = 0$$

$$\Rightarrow 8 + 4a + 2b + c = 0 \quad \ldots(2)$$

hence (1)-(2) gives

$$5a - 5b - 35 = 0$$

$$a - b = 7$$

$$P'(-3) = 27 - 6a + b < 0 \quad \ldots(3)$$

$$\Rightarrow 27 - 6(a - b) - 5b < 0$$

$$\Rightarrow 27 - 42 - 5b < 0$$

$$\Rightarrow -15 - 5b < 0$$

$$\Rightarrow 3 + b > 0$$

$$\Rightarrow b > -3, \text{ hence } a - 7 = b > -3$$

$$a - 7 > -3 \qquad [\text{from } (3)]$$

Q.52 Answer:

30

Solution:

$$\frac{4000}{a_1 a_{4001}} = 10 \Rightarrow a_1 a_{4001} = 400$$

$$a_1 + a_{4001} = 50$$

$$(a_1 - a_{4001})^2 = (a_1 + a_{4001})^2 - 4a_1 a_{4001}$$

$$\Rightarrow |a_1 - a_{4001}| = 30$$

Q.53 Answer:

$(1, 2, 3)$

Solution:

We have $\bar{r} = (11\hat{i} - 2\hat{j} - 8\hat{k}) + \lambda(10\hat{i} - 4\hat{j} - 11\hat{k})$

So coordinates of any point on this line are $[(10\lambda + 11), (-11\lambda - 2), (-11\lambda - 8)]$

Let $\mathrm{P} \equiv (2, -1, 5)$ and let $\underline{\mathrm{M}}$ be foot of perpendicular.

$\therefore$ d.r. of PM are $(10\lambda + 9), (-4\lambda - 1), (-11\lambda - 13)$

Sibce PM is perpendicular to given line, we write

$(10\lambda + 9)(10) + (4\lambda - 1)(-4) + (-11\lambda - 13)(-11) = 0$

$\therefore 100\lambda + 90 + 16\lambda + 4 + 121\lambda + 143 = 0 \Rightarrow 237 = -237\lambda$

$\Rightarrow \lambda = -1$

$M = -10 + 11, 4 - 2, 11 - 8)$ i.e. $(1, 2, 3)$

Q.54 Answer:

$\mathrm{M} + \mathrm{m} = 4(\pi + 1)$

Solution:

We have $f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t f(t)) dt$

$= \sin x + \pi \sin x + \int_{-\pi/2}^{\pi/2} t f(t) dt$

$\therefore \quad f(x) = (\pi + 1) \sin x + A \quad \ldots (1)$

Now $A = \int_{-\pi/2}^{\pi/2} t((\pi + 1) \sin t + A) dt$

$= 2(\pi + 1) \int_0^{\pi/2} \underset{(I)\ (II)}{t \sin t} \mathrm{dt}$

(Bypart)

$\Rightarrow \quad \mathrm{A} = 2(\pi + 1)$

Hence $\mathrm{f(x)} = (\pi + 1) \sin \mathrm{x} + 2(\pi + 1)$

$\therefore \quad \mathrm{f_{max.}} = 3(\pi + 1) = \mathrm{M}$

and $\mathrm{f_{min.}} = (\pi + 1) = \mathrm{m}$

$\Rightarrow \quad \frac{\mathrm{M}}{\mathrm{m}} = 3$

Q.55 Answer:

$\frac{27}{10}$

Solution:

$\frac{x^2}{9} + \frac{y^2}{1} = 1 \Rightarrow A(3, 0), B = (0, 1)$

The line $AB : \frac{x}{3} + \frac{y}{1} = 1$ meets the circle $x^2 + y^2 = 9$ at M

$$x^2+\left(1-\frac{x}{3}\right)^2=9 \Rightarrow \frac{10}{9}x^2-\frac{2x}{3}-8=0, 5x^2-3x-36=0$$

$$\Rightarrow x=-\frac{12}{5}, y=\frac{9}{5}$$

$$\therefore \quad M=\left(-\frac{12}{5}, \frac{9}{5}\right). \triangle AMO=\frac{1}{2}\begin{vmatrix} 3 & 0 & 1 \\ -\frac{12}{5} & \frac{9}{5} & 1 \\ 0 & 0 & 1 \end{vmatrix}=\frac{27}{10}.$$

Q.56 Answer:

$\frac{2}{9}$

Solution:

$$\begin{array}{ccccccc} \omega^{r_1} & + & \omega^{r_2} & + & \omega^{r_3} & = 0 \left\{1+\omega+\omega^2=0\right\} \\ r_1 & & r_2 & & r_3 & \\ \downarrow & & \downarrow & & \downarrow & \\ 3 \text{ or } 6 & & 1 \text{ or } 4 & & 2 \text{ or } 5 & \end{array}$$

required prob. $= \frac{2\times2\times2\times\underline{3}}{6^3} = \frac{2}{9}$

Q.57 Answer:

36

Solution:

$$\rightarrow f(x)=\cos^2 x+\sec^2 x+2$$

$$g(x)=\sin^2 x+\operatorname{cosec}^2 x+2$$

$$f(\pi/4)=g(\pi/4)$$

$$f(\pi/3)>g(\pi/3)$$

$$A=\int_{\pi/4}^{\pi/3}\cos^2 x+\sec^2 x-\sin^2 x-\operatorname{cosec}^2 x\cdot dx$$

$$=\int_{\pi/4}^{\pi/3}\cos 2x+\sec^2 x-\operatorname{cosec}^2 x\cdot dx$$

$$A=\frac{\sin 2x}{2}\Big|_{\pi/4}^{\pi/3}+\tan x+\cot x|_{\pi/4}^{\pi/3}$$

$$=\frac{\sqrt{3}}{4}-\frac{1}{2}+\sqrt{3}+\frac{1}{\sqrt{3}}-2$$

$$A=\frac{\sqrt{3}}{4}-\frac{5}{2}+\frac{4}{\sqrt{3}}$$

$$=\frac{3-10\sqrt{3}+16}{4\sqrt{3}}=\frac{19-10\sqrt{3}}{4\sqrt{3}}$$

$$a=19, b=10, c=3; d=4$$

$$a+b+c+d=36$$

Q.58 Answer:

2

Solution:

$z^2 - z + 1 = 0$

$\because z_1$ and z_2 are roots.

$\therefore z_1^2 - z_1 + 1 = 0$

and $z_2^2 - z_2 + 1 = 0$

Now, $z_1^4 - z_1^3 + 2z_1^2 - 2z_1 + 1$

$= z_1^2\left(z_1^2 - z_1 + 1\right) + \left(z_1^2 - z_1 + 1\right) - z_1 = -z_1$

Similarly, $z_2^4 - z_2^3 + 2z_2^2 - 2z_2 + 1$

$= z_2^2\left(z_2^2 - z_2 + 1\right) + \left(z_2^2 - z_2 + 1\right) - z_2$

$= -z_2$

$$\text{Hence } = -\left[\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{2013} + \left(\cos\frac{\pi}{3} - i\sin\pi/3\right)^{2013}\right]$$

$$= -\left[2\cos\left(\frac{2013\pi}{3}\right)\right]$$

$$E = 2$$

Q.59 Answer:

no value of λ and μ

Solution:

The given lines intersect, if the shortest distance between the lines is zero.

We know that the shortest distance between the lines $\mathbf{r} = \mathbf{a}_1 + \lambda\mathbf{b}_1$ and $\mathbf{r} = \mathbf{a}_2 + \mu\mathbf{b}_2$ is $\frac{|(\mathbf{a}_1 - \mathbf{a}_2)\cdot\mathbf{b}_1\times\mathbf{b}_2|}{|\mathbf{b}_1\times\mathbf{b}_2|}$

So the shortest distance between the given lines is zero if

$$(\mathbf{i} - \mathbf{j} - (2\mathbf{i} - \mathbf{j}))\cdot(2\mathbf{i} + \mathbf{k})\times(\mathbf{i} + \mathbf{j} - \mathbf{k}) = 0$$

$$= \begin{vmatrix} -1 & 0 & 0 \\ 2 & 0 & 1 \\ 1 & 1 & -1 \end{vmatrix} = 1 \neq 0$$

Hence the given lines do not intersect.

Q.60 Answer:

78.00

Solution:

$\text{correct } \Sigma x^2 = 2830 - 20^2 + 30^2 = 3330$

$\text{correct } \Sigma x = 170 - 20 + 30 = 180$

$\text{correct variance } = \frac{1}{15}\Sigma x^2 - \left(\frac{1}{15}\Sigma x\right)^2$

$= \frac{1}{15} \times 3330 - \left(\frac{1}{15} \times 180\right)^2$

$= 222 - 144$

$= 78$

Q.61 Answer:

8

Solution:

$f(\mathrm{x}) > 0$

$f(\mathrm{x}+\mathrm{y}+\mathrm{z}) = f(\mathrm{x}) \cdot f(\mathrm{y}) \cdot f(\mathrm{z}) \forall \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathrm{R} \quad \ldots(1)$

$\text{Put } \mathrm{x} = \mathrm{y} = \mathrm{z} = 0 \Rightarrow f(0) = 1$

$f'(2) = \lim_{\mathrm{h} \to 0^+} \frac{f(2+\mathrm{h}) - f(2)}{\mathrm{h}}$

Put $\mathrm{x} = 2, \mathrm{y} = \mathrm{h}, \mathrm{z} = 0$

$f(2+\mathrm{h}) = f(2) \cdot f(\mathrm{h}) \cdot f(0)$

$f(2+\mathrm{h}) = 4f(\mathrm{h})$

Put in (1)

$f'(2) = 4\lim_{h \to 0} \frac{f(h) - 1}{h}$

$= 4 \cdot f'(0) = 4.2 = 8$

Q.62 Answer:

0

Solution:

As we know $\sin^{-1} x$ **cannot be greater than** $\frac{\pi}{2}$**, So** $\sin^{-1} x = \sin^{-1} y = \sin^{-1} z = \frac{\pi}{2}$

Therefore $x = y = z = 1$

Putting these values in the expression, we get $1 + 1 + 1 - \frac{9}{1+1+1} = 0$

Q.63 Answer:

$\frac{x^2}{2y^2} \log\left(\frac{x}{y}\right) - \frac{x^2}{4y^2} = -\log y + \log c$

Solution:

$\frac{dx}{dy} = \frac{x^2 \log\left(\frac{x}{y}\right) - y^2}{xy \log\left(\frac{x}{y}\right)}$

Let $x = vy$ **and** $\frac{dx}{dy} = v + y\frac{dv}{dy}$

$$\therefore \quad v + y\frac{dv}{dy} = \frac{v^2 \log v - 1}{v \log v}$$

$$\Rightarrow v + y\frac{dv}{dy} = v - \frac{1}{v \log v}$$

$$y\frac{dv}{dy} = -\frac{1}{v \log v} \Rightarrow v \log v dv = -\frac{dy}{y}$$

On integration,

$$\int \log \mathrm{v} \cdot \mathrm{ddv} = -\int \frac{1}{\mathrm{y}} \mathrm{dy}$$

$$\Rightarrow (\log \mathrm{v})\frac{\mathrm{v}^2}{2} - \int \frac{1}{\mathrm{v}} \times \frac{\mathrm{v}^2}{2} \mathrm{dv} = -\log \mathrm{y} + \log \mathrm{c}$$

$$\Rightarrow \quad \frac{\mathrm{v}^2}{2} \log \mathrm{v} - \frac{\mathrm{v}^2}{4} = -\log \mathrm{y} + \log \mathrm{c}$$

$$\Rightarrow \frac{\mathrm{x}^2}{2\mathrm{y}^2} \log \left(\frac{\mathrm{x}}{\mathrm{y}}\right) - \frac{\mathrm{x}^2}{4\mathrm{y}^2} = -\log \mathrm{y} + \log \mathrm{c}$$

Q.64 Answer:

0

Solution:

$$L = \lim_{x \to 0} \left(\cos^2 x\right) \left(\int_0^{1/x} e^{t^2} dt\right)$$

$$1^{\infty} : L = e^{\lim_{x \to 0} \frac{-\int_0^{1/x} e^{t^2} dt}{\operatorname{cosec}^2 x}}$$

$$= \mathrm{e}^{-\lim_{x \to 0} \frac{\mathrm{e}^{\frac{1}{\mathrm{x}^2}} \left(-\frac{1}{\mathrm{x}^2}\right)}{2 \operatorname{cosec}^2 x \cot x}} = 0$$

Q.65 Answer:

7

Solution:

$$x + y + z = 51$$

$$x \geq 2, y \geq 2, z \geq 1$$

is equivalent to $\mathrm{x} + \mathrm{y} + \mathrm{z} = 46$

$\mathrm{x} \geq 0, \mathrm{y} \geq 0, \mathrm{z} \geq 0$

Now 46 can be divided is 23 pairs.

$$\Rightarrow \mathrm{x} + \mathrm{y} + \mathrm{z} = 23 \text{ pairs.}$$

$$\Rightarrow \text{ So number of solutions } = 23 + 3 - 1$$

$$= {}^{25}\mathrm{C}_2$$

Q.66 Answer:

0

Solution:

$$\left[{}^{10}C_0 + {}^{10}C_1 x^2 + {}^{10}C_2 x^4 + \ldots\right]\left(1 - x^2\right)^6$$
$$\Rightarrow \left[{}^{10}C_0 + {}^{10}C_1 x^2 + {}^{10}C_2 x^4 + \ldots\right]\left[1 - {}^{6}C_1 x^2 + {}^{6}C_2 x^4 \ldots\right]$$
$$\Rightarrow {}^{10}C_0 \cdot {}^{6}C_2 - {}^{10}C_1 \cdot {}^{6}C_1 + {}^{10}C_2$$
$$\Rightarrow 15 - 60 + 45 = 0$$

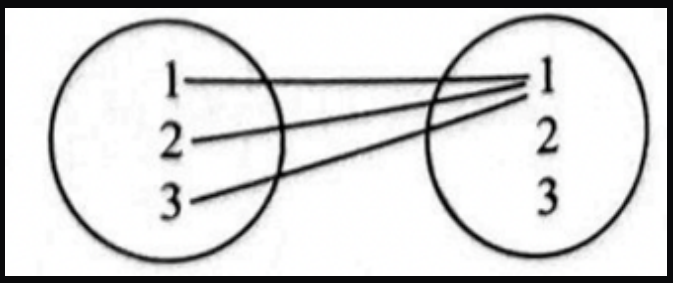
Q.67 Answer:

10

Solution:

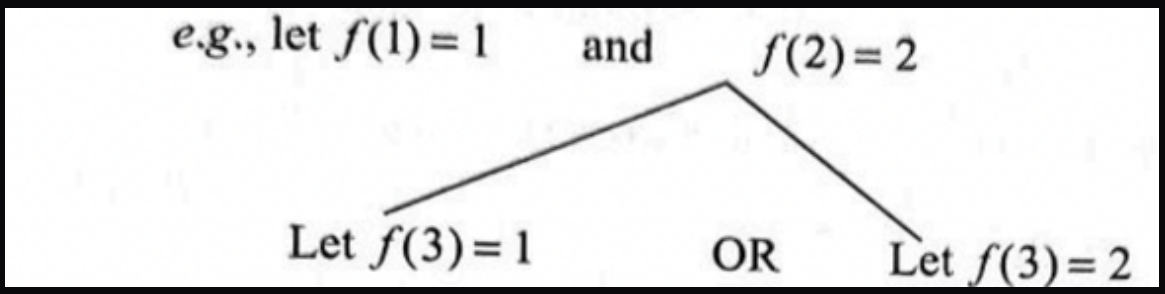
$f[f(x)] = f(x) \forall x \in S = \{1, 2, 3\}$

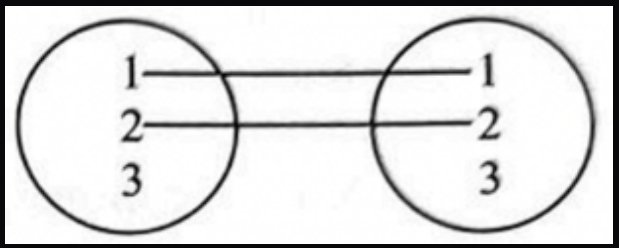
I. When range contains 1 element



${}^3C_1 \times 1 = 3$

II. When range contains 2 elements





$$\text{(i) If } x = 1, \quad \text{LHS} = f[f(x)] = f(1)$$
$$\text{RHS} = 1$$

In this case also LHS = RHS $\forall x \in S$

(ii) If $x = 2$, **LHS = RHS**

(iii) If $x = 3$, **LHS = RHS**

${}^3C_2 \times 2 = 6$

Remaining element can be mapped 2 ways.

III. When range contains 3 elements

$f(1) = 1 \quad f(2) = 2 \quad f(3) = 3$

Q.68 Answer:

4

Solution:

$$LHS = \begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix}$$

Operate: $C_1 \to C_1 + C_2 + C_3$

$$= \begin{vmatrix} 2(a+b+c) & a & b \\ 2(a+b+c) & b+c+2a & b \\ 2(a+b+c) & a & c+a+2b \end{vmatrix} = 2(a+b+c)\begin{vmatrix} 1 & a & b \\ 1 & b+c+2a & b \\ 1 & a & c+a+2b \end{vmatrix}$$

Operate: $R_2 \to R_2 - R_1; R_3 \to R_3 - R_1$

$$= 2(a+b+c)\begin{vmatrix} 1 & a & b \\ 0 & (a+b+c) & 0 \\ 0 & 0 & (c+a+b) \end{vmatrix}$$

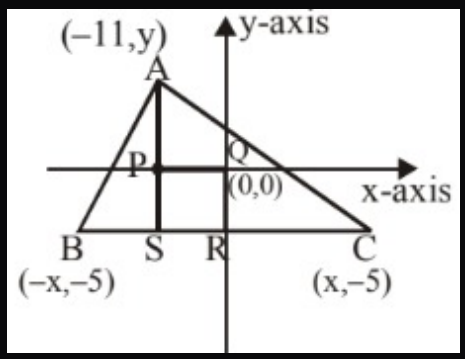
Expand by C_1

$$= 2(a+b+c).1 \cdot \begin{vmatrix} a+b+c & 0 \\ 0 & a+b+c \end{vmatrix} = 2(a+b+c)^3 \Rightarrow k = 2 \text{ and } \alpha = \beta = \gamma = 1$$

Q.69 Answer:

7

Solution:



Let $Q(0,0), P(-11,0), R(0,-5)$ **and** $S(-11,-5)$

Since R **is mid point of BC.**

Let $B(-x,-5)$ **then** C **is** $(x,-5)$.

Also $A(-11,y)$ **for some** y.

Now BP is an altitude, perpendicular to

$$AC \Rightarrow m_{BP} \cdot m_{AC} = -1$$

$$\text{or } \left(\frac{5}{x-11}\right)\left(\frac{y+5}{-11-x}\right) = -1$$

or $5(y+5) = (x+11)(x-11)$ **...(1)**

Also Q **is equidistant from** A & C,

so $y^2 + 121 = x^2 + 25$ **...(2)**

From (1) & (2), we get

$$5y + 25 = \left(y^2 + 96\right) - 121$$

or $y^2 - 5y - 50 = 0$ gives $y = 10, 5$

But $y = -5$ is not possible

Hence $y = 10 \Rightarrow x = 14$

$\therefore BC = 2x = 28$

Q.70 Answer:

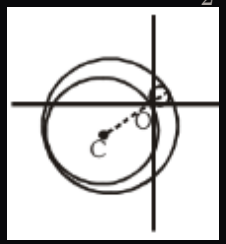
$\frac{3}{4}$

Solution:

$x^2 + y^2 + 4x + 6y - 3 = 0$

has centre $C(-2, -3)$ and radius $= 4$

$\therefore \quad r_2 = \frac{4+OC}{2} \& r_1 = \frac{4-OC}{2}$



$$\Rightarrow \quad r_2 = 2 + \frac{\sqrt{13}}{2} \ \& \ r_1 = 2 - \frac{\sqrt{13}}{2}$$

$$\therefore \quad r_1 r_2 = 4 - \frac{13}{4} = \frac{3}{4}$$

Q.71 Answer:

—

Solution:

$$A = \begin{vmatrix} 2 & 0 & -\alpha \\ 5 & \alpha & 0 \\ 0 & \alpha & 3 \end{vmatrix}; |A| = \begin{vmatrix} 2 & 0 & -\alpha \\ 5 & \alpha & 0 \\ 0 & \alpha & 3 \end{vmatrix} \neq 0$$

$6\alpha - 5\alpha^2 \neq = 0$

$\Rightarrow \alpha(6 - 5\alpha) \neq 0$

$\alpha = 0, 6/5$

$\therefore \quad \alpha \in R - \{0, 6/5\}$

For $\alpha = 1$

$$A = \begin{vmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{vmatrix} \Rightarrow |A| = 6 - 5 = 1; \text{Adj } A = \begin{vmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{vmatrix}$$

$$\therefore \quad A^{-1} = \begin{vmatrix} 3 & -1 & 1 \\ -15 & 6 & 5 \\ 5 & -2 & 2 \end{vmatrix}$$

By characteristic equation $|A - xI| = 0$

$$\begin{vmatrix} 2-x & 0 & -1 \\ 5 & 1-x & 0 \\ 0 & 1 & 3-x \end{vmatrix} = 0$$

$\Rightarrow x^3 - 6x^2 + 11x - 1 = 0$

By cayley hamilton theorem

$A^3 - 6A^2 + 11A = I \Rightarrow A^{-1} = A^2 - 6A + 11.I$

Q.72 Answer:

—

Solution:

$x^2 \dfrac{f'(g(x))g'(x)}{f(g(x))} = \left(1 - 2x^2\right) \cdot \dfrac{g'f(x) \cdot f'(x)}{g(f(x))}$

$x^2 \dfrac{d}{dx}(\ln f(g(x))) = \left(1 - 2x^2\right) \dfrac{d}{dx}(\ln(gf(x)))$

$\displaystyle\int_0^k fg(x)dx = 1 - e^{-k^2}$

$f(g(k)) = 2ke^{-k^2}$

$\ln f(g(x)) = \ln\left(2xe^{-x^2}\right)$

$\dfrac{d}{dx}\ln(fg(x)) = \dfrac{2e^{-x^2} - 4x^2e^{-x^2}}{2xe^{-x^2}} = \dfrac{2e^{-x^2}\left(-2x^2\right) + 1}{2e^{-x^2} \cdot x} = \dfrac{1}{x} - 2x$

Now,

$x^2 \dfrac{d}{dx}\ln(fg(x)) = \left(1 - 2x^2\right) \dfrac{d}{dx}(\ln gf(x))$

$\Rightarrow x^2 \cdot \dfrac{1 - 2x^2}{x} = \left(1 - 2x^2\right) \dfrac{d}{dx}\ln(gf(x))$

$\Rightarrow \ln g(f(x)) = \dfrac{x^2}{x} + c$

$\Rightarrow \ln\left(\dfrac{gf(3)}{gf(2)}\right) = \dfrac{9}{2} - 2 = \dfrac{5}{2}$

Q.73 Answer:

—

Solution:

$\lim_{n\to\infty} \tan^{-1}\left\{4n^2\left(1-\cos\frac{x}{n}\right)\right\} = \lim_{n\to\infty} \tan^{-1} \frac{\left\{4\left(1-\cos\frac{x}{n}\right)\cdot x^2\right\}}{\frac{x^2}{n^2}}$

$= \tan^{-1} 2x^2 \left\{\because \lim_{\theta\to 0} \frac{1-\cos\theta}{\theta^2} = \frac{1}{2}\right\}$

$\Rightarrow f(x) = \tan^{-1} 2x^2$

and $\lim_{n\to\infty} \frac{n^2}{2} \ell\cos\left(\frac{2x}{n}\right) = \lim_{n\to\infty} \frac{n^2}{2} \ln\left\{1+\cos\frac{2x}{n}-1\right\}$

$= \lim_{n\to\infty} \frac{n^2 \ell n\left\{1+\left(\cos\frac{2x}{n}-1\right)\right\}}{2\left(\cos\frac{2x}{n}-1\right)} \times \left(\cos\frac{2x}{n}-1\right)$

$= \frac{1}{2} \lim_{n\to\infty} \frac{\ln\left\{1+\left(\cos\frac{2x}{n}-1\right)\right\}}{\cos\frac{2x}{n}-1} \times \lim_{n\to\infty} \left(\frac{\cos\frac{2x}{n}-1}{\frac{4x^2}{n^2}}\right)\cdot 4x^2$

$\Rightarrow g(x) = -x^2 - 2g(x) \quad f(x)$

Now $L = \lim_{x\to 0} \frac{e^{-2g(x)}-e^{f(x)}}{x^6} = \lim_{x\to 0} \frac{e^{2x^2}-e^{\tan^{-1}2x^2}}{x^6}$

$\lim_{x\to 0} \frac{2x^2-\tan^{-1}2x^2}{x^6} = \frac{8}{3}$

Q.74 Answer:

—

Solution:

$x \to x+$

$5\begin{cases} f(x)+f(x+10)+\ldots+f(x+90) = f(x+5)+f(x+15)+\ldots+f(x+95) \\ f(x+5)+f(x+15)+\ldots+f(x+95) = f(x+10)+f(x+20)+\ldots+f(x+100) \end{cases}$

$f(x) = f(x+100)$

$f(x)$ **is periodic with period** $= 100$

$\int_0^{200} f(100-x)\cdot dx = 3 \Rightarrow \int_0^{100} f(100-x)\cdot dx = 3 \Rightarrow 2\int_0^{100} f(x)\cdot dx = 3\int_0^{100} f(x)\cdot dx = \frac{3}{2}$

$\int_0^{600} f(x)\cdot dx = 6\int_0^{100} f(x)\cdot dx = 6\times\frac{3}{2} = 9$

Q.75 Answer:

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Solution:

Let ' θ **' be the angle between** $\overrightarrow{\mathrm{a}}$ **and** $\overrightarrow{\mathrm{b}}$ **then**

$\because \quad |\vec{a}+\vec{b}|^2 = 1+1+2\cos\theta = 4\cos^2\theta/2$

and $\quad |\vec{a}-\vec{b}|^2 = 1+1-2\cos\theta = 4\sin^2\theta/2$

$\therefore \quad \frac{1}{|\vec{a}+\vec{b}|} + \frac{1}{|\vec{a}-\vec{b}|} = \frac{1}{4}\left(\operatorname{cosec}^2\theta/2 + \sec^2\theta/2\right)$

$= \frac{1}{4}\left(2+\tan^2\theta/2+\cot^2\theta/2\right)$

$\geq 1 \quad \{\text{using AM } \geq \text{GM}\}$